





*Research stay memoir*

# Random dynamical systems theory and application to turbulence

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# Random dynamical systems theory and application to turbulence

By Samuel Chan-ashing

**Abstract**

We study the chaotic motion of a turbulent system for which its flow is associated to a velocity field arising from the Navier-Stokes equations in 2D and 3D. Our approach involves exploiting randomness to consider the average behaviour of the fluid, and we thus subject the fluid to white-in-time, $H^s$-in-space stochastic forcing in a periodic box.

The key features of our study are tools from ergodic theory and random dynamical systems. The Multiplicative Ergodic Theorem allows us to define the Lyapunov exponent and a new version of Furstenberg's criterion, combined with hypoellipticity and approximate control arguments, allows us to prove the Lagrangian chaos of our system, that is, the top Lyapunov exponent is strictly positive.

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## 1. Introduction

### 1.1. Turbulence

In the physics and engineering literature, *turbulence* is used to describe a wide range of observed phenomena in dynamics of weakly dissipative, infinite dimensional and where the nonlinearity has to be of a certain type (fluids, nonlinear waves, etc). These conservative systems are described in *generic* settings, including both 2D and 3D incompressible or compressible Navier-Stokes, passive scalars advected by fluids, nonlinear dispersive equations, magneto-hydrodynamics models and kinetic models in plasmas.

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**Basic properties of turbulent systems**

All turbulent system shared some basic properties that are:

*Property of chaos.* The system is extremely sensitive to initial data.
Observation of chaos: if we subject the system to the same force and slightly change the initial data, we then get a very different answer (for example: exponentially fast).

*Property of ergodicity.* Time averages the same as running multiple experiments.

*Property of multiscale.* The system exhibits a multitude of scales and a cascade between scales: fills out an inertial range between the injection scale and the dissipation scale(s).
This forcing generally acts at sort of one set of scales, called *injection scale* (or *integral scale*), and the energy that it puts in, spreads out among all the other scales available in the system.
Observation of multiscale: if the previous operator N acts like a laplacian then what can sometimes happen is that the injection scale put in scale one, the nonlinearity mixing everythings up and sends information from low frequencies to high frequencies where the dissipation is very strong (even if $\nu \to 0$ the dissipation is never vanishing).

*Property of universality.* Statitics between the largest and smallest scales appear to be universal, *i.e.*, independant of details of experiment (e.g. basically independent of F and D).
Observation of universality: when we want to understand how the kinetic energy distribute itself through the inertial range we observe a graph, called the power spectrum, which is a log-log plot of the energy density as a function of frequency multiplied by $k$ to the dimension minus 1:

$$\mathrm{E}(k) \,=\, |k|^p \mathbb{E}(|\hat{u}(k)|^2).$$

*Property of anomalous dissipation.* A constant on-average flux of conserved quantities through the inertial range.

More generally, we have the following universal law:

| **Universal law of turbulence**. |
| --- |
| *Every turbulent systems are chaotic and ergodic.* |

**Observability and statistical stationarity**

Consider a fluid $u$. Most classical case is $t \to +\infty$ to statistical steady state, then focus on dissipation to zero. Thus, the equation studied has the general form:

$$\partial_t u \;=\; \mathrm{N}[u] \;+\; \nu \mathrm{D} u \;+\; \mathrm{F},$$

where N is an operator, $\nu$ is the dissipation and F is a external diving force.

Many turbulent systems are well-approximated as statistically stationary, like wind over a sail or water over a stone. For systems in fluid mechanics, we observe that these stationary statistics do not depend on the initial data and that time averages are the same as ensemble averages.

Hence, the main idea is that there should be an invariant measure $\mu$ over a phase space $\mathbf{H}$ such that if you randomize the initial condition (*i.e.*, $u_0 \overset{\text{Law}}{\sim} \mu$), then this law doesn't change as a function of time:

$$u_0 \overset{\text{Law}}{\sim} \mu \qquad \Longrightarrow \qquad \forall\, t \geqslant 0 \;:\quad u_t \overset{\text{Law}}{\sim} \mu.$$

NOTATIONS 1. For $f \in \mathrm{L}^1(\mathrm{X}, \mu)$, we denote:

$$\mathbb{E}_\mu(f) \,=\, \int_{\mathrm{X}} f(x)\, \mu(\mathrm{d}x) \qquad \text{and} \qquad \|f\|_{\mathrm{L}^p(\mathrm{X})} \,=\, \left(\mathbb{E}(|f|^p)\right)^{1/p}.$$

The *ensemble averages* of an observable $\varphi : \mathbf{H} \longrightarrow \mathbf{R}$ is given by

$$\mathbb{E}_\mu(\varphi) \;:=\; \int_{\mathbf{H}} \varphi(u)\,\mathrm{d}\mu(u).$$

However, we also want that the measure is *ergodic*: for $\mu$-a.e. initial condition $u$,

$$\lim_{\mathrm{T}+\infty} \frac{1}{\mathrm{T}} \int_0^{\mathrm{T}} \varphi(u_t)\,\mathrm{d}t \;=\; \mathbb{E}_\mu(\varphi).$$

More precisely, we want that these ergodic averages hold for *generic* data.

### 1.2. **Model**

Let $d = 2$ or 3. Stochastic differential equations model our real phenomenon well, because many turbulent systems are well-approximated by statistically stationary. However, the issue with this model is that for deterministic systems, the construction of the invariant measures which model this behavior is deeply challenging, even for relatively simple, finite dimensional ODEs. This provides a major first motivation for moving to random systems.

NOTATIONS 2. We use the following Fourier transform convention:

$$\hat{f}(k) \;=\; \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{T}^d} f(x) e^{-ik\cdot x}\,\mathrm{d}x.$$

We consider $(e_k)$ a real Fourier basis for functions on $\mathbb{T}^d$.
We denote $f \underset{a,b,c,\ldots}{\lesssim} g$ if there exists a constante $C = C(a,b,c,\ldots)$ such that $f \leqslant Cg$.

#### Phase space

The state space for our fluid velocity fields is $\mathrm{H}^s$, the Sobolev space of mappings with regularity $s$, with mean zero divergence free:

$$\mathbf{H} \;:=\; \left\{ u \in \mathrm{H}^s(\mathbb{T}^d, \mathbf{R}^d) \;:\quad \nabla \cdot u \;=\; 0 \quad \text{and} \quad \int_{\mathbb{T}^d} u \;=\; 0 \right\}.$$

We write: $\quad \|f\|_{\mathrm{H}^s} \;=\; \sum_{k \in \mathbf{Z}^d} |k|^{2s}\, |\hat{f}(k)|^2.$

#### Fluid mechanics models

We consider fluid mechanics subject to white-in-time, $\mathrm{H}^s$-in-space stochastic forcing in a periodic box $\mathbb{T}^d$ $(d = 2, 3)$ which is non-degenerate at high frequencies. We adopt an approach that considers the average behavior of fluids across all turbulent systems: the fluid velocity evolves according to a time-continuous, randomly driven ergodic motion. Then, let $u_t : \mathbb{T}^d \longrightarrow \mathbf{R}^d$ be the random velocity field at time $t > 0$ which evolves according to one of several incompressible, stochastically forced fluid mechanics models:

**System 1**. *2D Navier-Stokes system* in $\mathbb{T}^2$. This SPDE is:

$$\begin{cases} \partial_t u_t + u_t \cdot \nabla u_t \;=\; -\nabla p_t + \nu \Delta u_t + Q\dot{W}_t, \\ \nabla \cdot u_t \;=\; 0, \\ u_0 \in \mathbf{H}. \end{cases}$$

**System 2**. *3D hyper-viscous Navier-Stokes system* in $\mathbb{T}^3$. This SPDE is:

$$\begin{cases} \partial_t u_t + u_t \cdot \nabla u_t \;=\; -\nabla p_t + \nu \Delta u_t - \eta \Delta^2 u_t + Q\dot{W}_t, \\ \nabla \cdot u_t \;=\; 0, \\ u_0 \in \mathbf{H}. \end{cases}$$

where:

- $W_t$: is a *cylindrical Wiener process* on $\mathbf{H}$ defined by

$$W_t := \sum_{m \in \mathbf{Z}^d \setminus \{0\}} e_m \gamma_m W_t^m,$$

where $(W_t^m)_m$ is a family of independent $(d-1)$-dimensional Wiener processes on a common canonical filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$.

- $Q$: is a *positive Hilbert-Schmidt operator* on mean-zero, divergence free $\mathrm{L}^2$ vector fields with singular values $(q_k)_{k \in \mathbf{Z}^d \setminus \{0\}}$ satisfying:

$$q_k \approx |k|^{-\alpha},$$

for an arbitrary, fixed $\alpha > 5d/2$. This condition says that the forcing is supported in every mode (we can also only consider the case of high frequencies $q_k \gtrsim |k|^{-\alpha}$).

For the remainder of this article, we take $s \in \left(\alpha - 2(d-1), \alpha - \frac{d}{2}\right)$ and thus $s > \frac{d}{2} + 3$, which implies that the velocities are always at least $\mathcal{C}^3$ in space:

$$\mathbf{H} \hookrightarrow \mathcal{C}^3.$$

**Almost-sure global well-posedness**

Let's re-write the Navier-Stokes system as an evolution equation on $\mathbf{H}$:

$$\partial_t u + \mathrm{B}(u,u) + \mathrm{A}u = Q\dot{W} = \sum_{m \in \mathbf{Z}^d \setminus \{0\}} q_m e_m \dot{W}^m,$$

where the Euler nonlinearity in both 2D and 3D is denoted by:

- $\mathrm{B}(u,v) = \nabla \cdot p(u \otimes v)$,
- $p = (\mathrm{id} - \nabla(-\Delta)^{-1}\nabla\cdot)$ is the Leray projection,
- $\mathrm{A}u = \begin{cases} -\nu\Delta u & \text{if } d = 2, \\ -\nu\Delta u + \eta\Delta^2 u & \text{if } d = 3. \end{cases}$ is the dissipation operator.

*Remark.* Note that: $e^{-t\mathrm{A}}$ is the propagator of the heat semigroup in 2D, and $\eta$ is the hyperviscosity in 3D.

We follow the standard procedure of defining the $(u_t)$ process to be a solution of equation in the mild sens, *i.e.*,

**Proposition 1** (Mild formulation)**.** *The $(u_t)$ process with initial data $u$ is a solution of:*

$$u_t = e^{-t\mathrm{A}}u + \Gamma_t + \int_0^t e^{-(t-s)\mathrm{A}}\mathrm{B}(u_s, u_s)\,\mathrm{d}s,$$

*where $\Gamma_t = \displaystyle\int_0^t e^{-(t-s)A} Q\,\mathrm{d}W(s)$ is the pertinent stochastic convolution for ou additive noise, in 2D.*

*Remark.* In 3D, solutions could break down in finite time (we don't know), and we will need to consider the stochastic analogue of Leray-Hopf weak solutions to send $t \longrightarrow +\infty$.

For the mild formulation, we have the following well-posedness theorem.

**Theorem 2** (Almost-sure global well-posedness)**.** *We have that:*

$$\mathbb{P}\text{-}\,\forall \exists\, \omega \in \Omega, \quad \forall\, p \geqslant 1, \quad \forall\, T > 0, \quad \forall\, \gamma < \alpha - \frac{d}{2}, \quad \forall\, u \in \mathbf{H} \cap \mathrm{H}^\gamma,$$

*there exists a unique solution $(u_t)$ to the mild formulation. Moreover, we have that:*

$(i)$ $(u_t)$ *is* $\mathcal{F}_t$*-adapted,*

$(ii)$ $u \in \mathrm{L}^p\left(\Omega,\, \mathcal{C}^\circ([0,T],\, \mathbf{H}\cap \mathrm{H}^\gamma)\right) \,\cap\, \mathrm{L}^2\left(\Omega,\, \mathrm{L}^2((0,T),\, \mathrm{H}^{\gamma+(d-1)})\right)$.

$(iii)$ $\forall\, p \geqslant 1, \quad \forall\, \beta \in (\gamma,\, \alpha - \frac{d}{2})$ :

$$\begin{aligned}
\mathbb{E}_W\left(\sup_{t\in[0,T]} \|u_t\|^p_{\mathrm{H}^\gamma}\right) &\underset{T,p,\gamma}{\lesssim} 1 + \|u\|^p_{\mathbf{H}\cap\mathrm{H}^\gamma}\,,\\
\mathbb{E}_W\left(\int_0^T \|u_s\|^2_{\mathrm{H}^{\gamma+(d-1)}}\,\mathrm{d}s\right) &\underset{T,\delta}{\lesssim} 1 + \|u\|^2_{\mathrm{H}^\gamma}\,,\\
\mathbb{E}_W\left(\sup_{t\in[0,T]}\left(t^{\frac{\beta-\gamma}{2(d-1)}}\,\|u_t\|_{\mathrm{H}^\beta}\right)^p\right) &\underset{p,T,\gamma,\beta}{\lesssim} 1 + \|u\|^p_{\mathrm{H}^\gamma}\,.
\end{aligned}$$

*Proof in 2D.* The proof is based on mild formulation, regularization of heat semigroup, and properties of stochastic convolution $\Gamma_t$. Estimates on $\Gamma_t$ follows from the Factorization Lemma, the Burkholder-Davis-Gundy Inequality, and the smoothing properties of the heat semigroup. □

## 2. **Random dynamical systems theory**

In this paper we formalize the notion of a random dynamical system on a measurable set X. An initial motivation for this formalism is to model the iteration of a dynamical system subjected to a random noise. Nevertheless, motivated in particular by fundamental mathematical questions, the general framework has largely deviated from this initial idea and makes it possible to encompass much more varied situations. Let:

- $(\mathrm{T}, +)$ be a semi-group,
- $(\mathrm{X}, d)$ be a separable, complete and metric space,
- $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space such that $\Omega$ is Borel subset of a Polish space and $\mathcal{F}$ is the set of Borel subsets of $\Omega$.

### 2.1. **Random dynamical systems**

**Definition 1.** We say that $\theta : \Omega^{\mathrm{T}} \longrightarrow \Omega^{\mathrm{T}}$ is a **semiflow** on $\Omega$ if:

$(i)$ $\left|\begin{array}{rccl} \theta: & \Omega\times\mathrm{T} & \longrightarrow & \Omega,\\ & (\omega,t) & \longmapsto & \theta^t\omega, \end{array}\right.$ is a measurable mapping,

$(ii)$ $\forall\,\omega\in\Omega\ :\quad \theta^0\omega = \omega,$

$(iii)$ $\forall\, s,t\in\mathrm{T}\ :\quad \theta^t\circ\theta^s = \theta^{t+s}.$

**Definition 2.** We say that $(\theta^t)$ is a **measure-preserving semiflow** on $\Omega$ if $(\theta^t)$ is a semiflow such that:

$$\forall\, t\in\mathrm{T}\ :\quad (\theta^t)^*\mathbb{P} = \mathbb{P},$$

where $f^*\mu = \mu(f^{-1}(\cdot))$. We then call $(\Omega, \mathcal{F}, \mathbb{P}, (\theta^t)_{t\in\mathrm{T}})$ a *measure-preserving system* (or *dynamical system*, abbreviated DS).

**Definition 3.** We say that $\mathrm{A}\subset\mathrm{X}$ is a **full $\mu$-measure set** if $\mu(\mathrm{X}\backslash\mathrm{A}) = 0$.

Notations 3. For a borel positive measure $\mu$ on X we denote its *support* as:

$$\mathrm{supp}\,(\mu) \ :=\ \bigcap_{\substack{\mathrm{F\ closed}\\ \mu(\mathrm{X}\backslash\mathrm{F})=0}} \mathrm{F},$$

the intersection of all full $\mu$-measure closed sets.

**Definition 4.** We call **random dynamical system** (abbreviated RDS) on a measurable space $(\mathrm{X}, \mathcal{B}_\mathrm{X})$ over $(\Omega, \mathcal{F}, \mathbb{P}, (\theta^t)_t)$ a family of mappings $(\mathcal{T}_\omega)_{\omega\in\Omega}$ defined for all $\omega \in \Omega$ by:

$$\begin{array}{rccc} \mathcal{T}_\omega : & \mathrm{T}\times\mathrm{X} & \longrightarrow & \mathrm{X}, \\ & (t,x) & \longmapsto & \mathcal{T}_\omega^t(x), \end{array}$$

and satisfying:

($i$) (measurability):

$$\begin{array}{rccc} \mathcal{T} : & \Omega\times\mathrm{T}\times\mathrm{X} & \longrightarrow & \mathrm{X}, \\ & (\omega,t,x) & \longmapsto & \mathcal{T}_\omega^t(x), \end{array} \quad \text{is } (\mathcal{F}\otimes\mathcal{B}_\mathrm{T}\otimes\mathcal{B}_\mathrm{X}, \mathcal{B}_\mathrm{X})\text{-mesurable,}$$

($ii$) (cocyle property): $\mathcal{T}_\omega^t$ form a cocycle over $\theta$

$$\forall\,\omega\in\Omega, \quad \forall\, s,t\in\mathrm{T} \ : \quad \mathcal{T}_\omega^{t+s} = \mathcal{T}_{(\theta^s\omega)}^t \circ \mathcal{T}_\omega^s,$$

and $\mathcal{T}_\omega^0 = \mathrm{id}_\mathrm{X}$ if $0 \in \mathrm{T}$ (for all $\omega \in \Omega$).

*Remark.* The composition creates a certain break of symmetry in the theory.

**Definition 5.** When $\mathrm{T} = \mathbf{Z}_+$, the phase $\mathcal{T}_\omega^n(x)$ represents the state of the system after $n$ iterations. In this case, we call **discrete time RDS**. When $\mathrm{T} = \mathbf{R}_+$, the phase $\mathcal{T}^t(x)$ represents the state of the system at time $t$ from $x$ at $t = 0$. In this case, we call **continuous time RDS**.

### Smoothness properties

Let $\mathcal{T}$ be a RDS on X.

**Definition 6.** $\mathcal{T}$ is said to be:

- **continuous** (or *topological*) on X if for all $\omega \in \Omega$ : $\mathcal{T}_\omega$ is continuous on $\mathrm{T}\times\mathrm{X}$.
- **smooth of class** $\mathcal{C}^k$ if for all $(\omega, t) \in \Omega\times\mathrm{T}$ : $\mathcal{T}_\omega^t$ is smooth of class $\mathcal{C}^k$ on X.
- **linear** if X is a finite-dimensional vector space and for all $(\omega, t) \in \Omega\times\mathrm{T}$ : $\mathcal{T}_\omega^t \in \mathcal{L}(\mathrm{X})$.

We introduce a stronger definition of continuity for our system as follows.

Notations 4. For metric spaces V, W, we define the space of strongly continuous maps $\mathcal{C}_{u,b}(\mathrm{V},\mathrm{W})$ as the space of continuous function $f : \mathrm{V} \longrightarrow \mathrm{W}$ such that for all $\mathrm{B} \subset \mathrm{V}$ bounded:

($i$) $f_{|\mathrm{B}}$ is uniformly continuous on B,

($ii$) $f(\mathrm{B})$ is bounded.

Hence we say that:

**Definition 7.** $\mathcal{T}$ on X is **strongly continuous** on $\mathrm{T}\times\mathrm{X}$ if for all $\omega \in \Omega$ : $\mathcal{T}_\omega \in \mathcal{C}_{u,b}(\mathrm{T}\times\mathrm{X}, \mathrm{X})$.

### Increments condition

For the remainder of our paper, we assume that the semigroup $\mathrm{T} = \mathbf{R}_+$.

**Definition 8.** For $s \leqslant t \in \mathrm{T}$, we call a **increment** the following random variable:

$$\mathcal{T}_{\theta^s(\cdot)}^{t-s} \ : \ \Omega \ \longrightarrow \ \mathrm{X}^\mathrm{X}.$$

**Definition 9.** We say that $\mathcal{T}$ has **independent increments** if:

$$\forall\, s,t\in\mathrm{T} \ : \quad \mathcal{T}_\omega^t \text{ is independent of } \mathcal{T}_{\theta^t\omega}^s.$$

**Proposition 3** (Characterisation)**.** *$\mathcal{T}$ has independent increments if and only if:*

$$\forall\, t_0 \leqslant t_1 \leqslant \ldots \leqslant t_n \;:\quad \mathcal{T}^{t_1-t_0}_{\theta^{t_0}(\cdot)},\ \mathcal{T}^{t_2-t_1}_{\theta^{t_1}(\cdot)},\ \ldots\ ,\mathcal{T}^{t_n-t_{n-1}}_{\theta^{t_{n-1}}(\cdot)} \text{ are independent.}$$

**Theorem 4** (Stationary increments)**.** *Each RDS with independent increments has stationary increments and:*

$$\forall\, t \geqslant s \;:\quad \mathbb{P}_{(\mathcal{T}^{t-s}_{\theta^s(\cdot)})} = \mathbb{P}_{(\mathcal{T}^{t-s}_{(\cdot)})}.$$

*In particular, by the $\theta^t$ invariance of $\mathbb{P}$ :*

$$\forall\, s \in \mathrm{T} \;:\quad \mathcal{T}^t_{\theta^s\omega} = \mathcal{T}^t_\omega$$

*Proof.* The $\theta^t$-invariance of $\mathbb{P}$ implies that the joint distribution of the increments doesn't change if the $t_i$ are replaced by $t_i + t \in \mathrm{T}$. □

## 2.2. **Markov RDS**

Let $(x_t)_{t\geqslant 0}$ be a stochastic process.

**Definition 10.** Let $x \in \mathrm{X}$ be the initial point of $(x_t)$. We say that $(x_t)_{t\geqslant 0}$ is **derived from** RDS $\mathcal{T}$, if:

$$\begin{cases} x_t = \mathcal{T}^t_\omega x_0, \\ x_0 = x. \end{cases}$$

**Lemma 5.** *If $\mathcal{T}$ is continuous and has independent increments, then $(x_t)_{t\geqslant 0}$ is Markovian.*

*Proof.* Straightforward. □

We extend the transition function of a stochastic process to a RDS:

**Definition 11.** We associate the **Markov semi-group** (or *Markov kernel*) to $\mathcal{T}$ as the operators defined for all $t \in \mathrm{T}$ as:

$$\begin{array}{rccl} \mathrm{P}_t: & \mathrm{X}\times\mathcal{B}_\mathrm{X} & \longrightarrow & \mathbf{R} \\ & (x,\ \mathrm{K}) & \longmapsto & \mathbb{P}\left(\left\{\omega\in\Omega\ :\ \mathcal{T}^t_\omega x \in \mathrm{K}\right\}\right). \end{array}$$

The Markov property of $(x_t)$ implies the semi-group relation: $\mathrm{P}_{s+t} = \mathrm{P}_t \circ \mathrm{P}_s$.

*Remark.* We can rewrite the Markov semi-group as:

$$\mathrm{P}_t(x,\mathrm{K}) = \int_\Omega \mathbb{1}_\mathrm{K}(\mathcal{T}^t_\omega x)\,\mathrm{d}\mathbb{P}(\omega).$$

To each Markov kernel there correspond two families of linear operators acting in the spaces of bounded measurable functions $\mathrm{L}^\infty(\mathrm{X})$ and of probability measures $\mathcal{P}(\mathrm{X})$.

**Definition 12.** The **transition operator** $\mathrm{P}_t : \mathrm{L}^\infty(\mathrm{X}) \longrightarrow \mathrm{L}^\infty(\mathrm{X})$ of $\mathcal{T}$ is defined for all $h \in \mathrm{L}^\infty(\mathrm{X},\mathbf{R})$ by:

$$\begin{array}{rccl} \mathrm{P}_t h: & \mathrm{X} & \longrightarrow & \mathbf{R} \\ & x & \longmapsto & \displaystyle\int_\mathrm{X} h(y)\mathrm{P}_t(x,\mathrm{d}y). \end{array}$$

*Remark.* Note that for all $\mathrm{K} \in \mathcal{B}_\mathrm{X}$, we have:

$$\mathrm{P}_t\mathbb{1}_\mathrm{K}(x) = \int_\mathrm{X} \mathbb{1}_\mathrm{K}(y)\mathrm{P}_t(x,\mathrm{d}y) = \int_\mathrm{K} \mathrm{P}_t(x,\mathrm{d}y) = \mathrm{P}_t(x,\mathrm{K}).$$

**Proposition 6.** *We have:*

$$\mathrm{P}_t h(x) = \int_\Omega h(\mathcal{T}^t_\omega x)\,\mathrm{d}\mathbb{P}(\omega).$$

*Proof.* By the previous remark, for all $\mathrm{K} \in \mathcal{B}_{\mathrm{X}}$ we have:

$$\mathrm{P}_t \mathbb{1}_{\mathrm{K}}(x) \;=\; \mathrm{P}_t(x, \mathrm{K}) \;=\; \int_\Omega \mathbb{1}_{\mathrm{K}}(\mathcal{T}_\omega^t x)\, \mathrm{d}\mathbb{P}(\omega).$$

By linearity, and then by the Beppo Levi theorem, we conclude that the result holds for all bounded measurable function $h$. □

**Definition 13.** We define the **dual** $(\mathrm{P}_t^*)$ of the operator $(\mathrm{P}_t)$ on the space of finite signed Borel measures on X such as for each $\mu$:

$$\begin{array}{rccc} \mathrm{P}_t^*\mu : & \mathcal{B}_{\mathrm{X}} & \longrightarrow & \mathbf{R} \\ & \mathrm{K} & \longmapsto & \displaystyle\int_{\mathrm{X}} \mathrm{P}_t(x, \mathrm{K})\, \mathrm{d}\mu(x). \end{array}$$

**Smoothness properties**

**Definition 14** (Feller property)**.** We say that $(\mathrm{P}_t)$ is **Feller** if for all $t \in \mathrm{T}$:

$$\forall\, h \in \mathcal{C}_b(\mathrm{X}, \mathbf{R}) \;:\quad \mathrm{P}_t h \;\text{ is continuous on X.}$$

*Remark.* Of course this implies that each $\mathrm{P}_t h$ is defined on X.

**Definition 15** (Continuity)**.** We say that the semigroup $(\mathrm{P}_t)$ is **continuous on** $\mathcal{C}_{u,b}(\mathrm{X}, \mathbf{R})$ if for all $h \in \mathcal{C}_{u,b}(\mathrm{X}, \mathbf{R})$:

- $\forall\, t > 0 \;:\; \mathrm{P}_t h \in \mathcal{C}_{u,b}(\mathrm{X}, \mathbf{R})$,
- $t \geqslant 0 \longmapsto \mathrm{P}_t h$ is continuous in the topology on $\mathcal{C}_{u,b}(\mathrm{X}, \mathbf{R})$.

**Proposition 7.** *Let $\mathcal{T}$ be a strongly continuous RDS. Then:*

$$\mathcal{T} \text{ has independent increments} \quad\Longrightarrow\quad (\mathrm{P}_t) \text{ is Feller and continuous on } \mathcal{C}_{u,b}(\mathrm{X}, \mathbf{R}).$$

*Proof.* Straightforward. □

## 2.3. **Invariant and stationary measures**

Let:

- $(\mathrm{X}, d)$ be a separable, complete and metric space,
- $(\Omega, \mathcal{F}, \mathbb{P}, (\theta^t)_{t\in\mathrm{T}})$ be a measure-preserving system,
- $\mathcal{T}$ be a strongly continuous RDS with independent increments and $(\mathrm{P}_t)$ its Markov kernel.

**Skew product and invariant measures**

The Markov chain formulation given above is useful in that it identifies stationary measures, *i.e.*, time-invariant statistics on X for the RDS. However, the Markov Kernel loses some structure of the RDS in the sens that: Markov kernel can arise from qualitatively different RDS.

We can encodes the entire RDS by extending $\theta$ by $\mathcal{T}$ as a single "deterministic", measurable semiflow on the augmented space $\Omega \times \mathrm{X}$:

**Definition 16.** The **skew product** associated to $\mathcal{T}$ is the mapping

$$\begin{array}{rccc} \Theta : & \Omega \times \mathrm{T} \times \mathrm{X} & \longrightarrow & \Omega \times \mathrm{X} \\ & (\omega, t, x) & \longmapsto & \Theta^t(\omega, x) \;=\; (\theta^t\omega, \mathcal{T}_\omega^t x), \end{array}$$

where:

$$\forall\, t \in \mathrm{T} \;:\quad \Theta^t \;\text{ is a measurable DS on } (\Omega \times \mathrm{X},\; \mathcal{F} \otimes \mathcal{B}_{\mathrm{X}})$$

*Remark.* Conversely, every such measurable skew product DS $\Theta$ defines a cocycle on its $x$ component, thus a measurable RDS. We can thus use RDS $\mathcal{T}$, cocycle $\mathcal{T}$ and skew product $\Theta$ synonymously.

We can thus see that the study of our random dynamical system is reduced to that of the deterministic dynamic system $(\Omega^{\mathrm{T}} \times \mathrm{X}, \mathbb{P})$.

**Definition 17.** We define the **dual** $((\Theta^t)^*)$ of the skew product $(\Theta^t)$ on the space of finite signed Borel measures on $\Omega \times \mathrm{X}$ such as for each $\eta$:

$$(\Theta^t)^* \eta \,=\, \eta\left((\Theta^t)^{-1}(\cdot)\right)$$

The skew product provides us a connection between standard ergodic theory, *i.e.*, the theory of invariant measures for individual mappings of a measurable space, and our present setting of random dynamical systems.

**Definition 18.** We say that $\eta$ is a **invariant measure** on $\Omega \times \mathrm{X}$, if $\eta$ is a probability measure such that for the semiflow $\Theta$:

$$\forall\, t \in \mathrm{T} \quad : \quad (\Theta^t)^* \eta \,=\, \eta.$$

**Stationary measures**

Consider a Borel probability $\mu$ on $(\mathrm{X}, \mathcal{B}_{\mathrm{X}})$ and recall that $(\mathrm{P}_t)$ is the Markov kernel of $\mathcal{T}$.

**Definition 19.** We say that the Borel probability $\mu$ on X is a **stationary measure** for $\mathcal{T}$ if:

$$\forall\, t \in \mathrm{T} \,:\quad \mathrm{P}_t^* \mu \,=\, \mu.$$

**Proposition 8** (Characterisation)**.** *We have that $\mu$ is a stationary measure for $\mathcal{T}$ if and only if:*

$$\forall\, f \in \mathrm{L}^1(\mu),\ f \,\geqslant\, 0 \,:\quad \int_{\mathrm{X}} \mathrm{P}_t f(x) \mu(\mathrm{d}x) \,=\, \int_{\mathrm{X}} f(x) \mu(\mathrm{d}x).$$

**Lemma 9.** *Let $\mathcal{T}$ be a strongly continuous RDS with independent increments. Then:*

- *if* X *is compact, or*
- *if there exists a Borel probability $\mu_0$ for which the sequence $(\mathrm{P}_t)^* \mu_0$ is tight,*

*then $\mathcal{T}$ admits at least one stationary measure $\mu$.*

*Proof.* It's a consequence of a standard Krylov-Bogoliubov argument. □

The following theorem makes this connection between standard ergodic theory and random dynamical systems:

**Theorem 10** (Connection between SET and RDS, Ohno [24])**.** *Assume $\mathcal{T}$ is a strongly continuous RDS with independent increments. Let $\mu$ be a Borel probability measure on* X*. Then:*

$$\mathbb{P} \times \mu \ \text{ is invariant for } (\Theta^t) \qquad \Longleftrightarrow \qquad \mu \ \text{ is stationary for } (\mathrm{P}_t).$$

*Proof.* Let $t \in \mathrm{T}$ and $\varphi \in \mathrm{L}^\infty(\Omega \times \mathrm{X})$. We denote for all $x \in \mathrm{X}$:

$$\widehat{\varphi}(x) \,:=\, \int_{\Omega} \varphi(\omega, x) \mathrm{d}\mathbb{P}(\omega).$$

Then:

$$\mathbb{E}_{\Theta^t \#(\mathbb{P} \times \mu)}(\varphi) \,=\, \int \varphi \,\mathrm{d}\left(\Theta^t \#(\mathbb{P} \times \mu)\right) \,=\, \int_{\Omega \times \mathrm{X}} \varphi\left(\Theta^t(\omega, x)\right) \mathrm{d}(\mathbb{P} \times \mu)(\omega, x).$$

However, since $\mathcal{T}$ has independent increments we have:

$$\begin{aligned}\int_{\mathrm{X}}\int_{\Omega}\varphi\left(\theta^t\omega,\mathcal{T}_\omega^t x\right)\mathrm{d}\mathbb{P}(\omega)\mathrm{d}\mu(x) &= \int_{\mathrm{X}}\int_{\Omega}\left(\int_{\Omega}\varphi\left(\tilde{\omega},\mathcal{T}_\omega^t x\right)\mathrm{d}\mathbb{P}(\tilde{\omega})\right)\mathrm{d}\mathbb{P}(\omega)\mathrm{d}\mu(x)\\ &= \int_{\mathrm{X}}\left(\int_{\Omega}\widehat{\varphi}(\mathcal{T}_\omega^t x)\mathrm{d}\mathbb{P}(\omega)\right)\mathrm{d}\mu(x).\end{aligned}$$

Or the Proposition 6 gives us that:

$$\int_{\Omega}\widehat{\varphi}(\mathcal{T}_\omega^t x)\mathrm{d}\mathbb{P}(\omega) = \mathrm{P}_t\widehat{\varphi}(x).$$

Thus we have by definition of $\mathrm{P}_t\widehat{\varphi}$ and integrating with respect to $\mathrm{d}\mu$:

$$\int_{\mathrm{X}}\mathrm{P}_t\widehat{\varphi}(x)\mathrm{d}\mu(x) = \iint_{\mathrm{X}\times\mathrm{X}}\widehat{\varphi}(y)\left(\mathrm{P}_t(x,\mathrm{d}y)\mathrm{d}\mu(x)\right) = \int_{\mathrm{X}}\widehat{\varphi}(y)\,\mathrm{d}(\mathrm{P}_t^*\mu)(y) = \mathbb{E}_{\mathrm{P}_t^*\mu}(\widehat{\varphi}).$$

Hence we have:

$$\forall\,\varphi\in\mathrm{L}^\infty(\Omega\times\mathrm{X}) \;:\quad \mathbb{E}_{\Theta^t\#(\mathbb{P}\times\mu)}(\varphi) = \mathbb{E}_{\mathrm{P}_t^*\mu}(\widehat{\varphi}).$$

Finally, such that:

$$\mathbb{E}_\mu(\widehat{\varphi}) = \int_{\mathrm{X}}\widehat{\varphi}(x)\mathrm{d}\mu(x) = \int_{\mathrm{X}}\int_{\Omega}\varphi(\omega,x)\mathrm{d}\mathbb{P}(\omega)\mathrm{d}\mu(x) = \int_{\Omega\times\mathrm{X}}\varphi(\omega,x)\mathrm{d}(\mathbb{P}\times\mu) = \mathbb{E}_{\mathbb{P}\times\mu}(\varphi),$$

we have $(\mathrm{P}_t^*\mu) = \mu$ if and only if $\mathbb{P}\times\mu$ is $\Theta$-invariant. □

**Ergodic measures**

Let $\mu$ be a stationary measure for $\mathcal{T}$.

**Definition 20.** We say that:

- $h\in\mathrm{L}^\infty(\mathrm{X})$ is $(\mathcal{T},\mu)$**-invariant function** if:

$$\forall\, t\geqslant 0 \;:\quad \mathrm{P}_t h = h \qquad \mu\text{-a.s.}$$

- $\mathrm{K}\subset\mathrm{X}$ is $(\mathcal{T},\mu)$**-invariant set** if:

$$\mathbb{1}_{\mathrm{K}} \text{ is } (\mathcal{T},\mu)\text{-invariant.}$$

**Definition 21.** We say that a stationary measure $\mu$ is **ergodic** if the only $(\mathcal{T},\mu)$-invariant functions are $\mu$-a.s constant:

$$\forall\, t\geqslant 0 \;:\quad \mathrm{P}_t\varphi = \varphi \quad \mu\text{-a.s} \qquad\Longrightarrow\qquad \varphi = c\in\mathbf{R} \quad \mu\text{-a.s.}$$

**Proposition 11.** *Let $\mu$ be a stationary measure for $\mathcal{T}$. Then:*

$$\mathbb{P}\times\mu \text{ is ergodic for } (\Theta^t) \qquad\Longleftrightarrow\qquad \mu \text{ is ergodic for } (\mathrm{P}_t).$$

*Proof.* This proposition is proved in [24]. □

## 3. Lyapunov exponent

We introduce the notion of linear cocycle over a given RDS and formulates the Multiplicative Ergodic Theorem (MET) allowing us to define the Lyapunov exponent. Let:

- $(\mathrm{X},d)$ be a separable, complete and metric space,
- $(\Omega,\mathcal{F},\mathbb{P},(\theta^t)_{t\in\mathrm{T}})$ be a measure-preserving system,
- $\mathcal{T}$ be a strongly continuous RDS on X referred to below as the *base* RDS, and $(\mathrm{P}_t)$ its Markov kernel.

### 3.1. **Materials**

NOTATIONS 5. For a matrix $A \in \mathcal{M}_d(\mathbf{R})$, we write $\sigma_i(A)$ as the $i$-th singular value of $A$.

**Definition 22.** A **linear cocycle** $\mathcal{A}$ over $\mathcal{T}$ is a mapping:

$$\mathcal{A} \,:\, \Omega \times \mathrm{T} \times \mathbf{R}^d \;\longrightarrow\; \mathcal{M}_d(\mathbf{R}).$$

with the following properties:

($i$) the evaluation mapping

$$\begin{array}{ccc} \Omega \times \mathrm{T} \times \mathrm{X} & \longrightarrow & \mathcal{M}_d(\mathbf{R}), \\ (\omega, t, x) & \longmapsto & \mathcal{A}^t_{\omega,x}, \end{array} \quad \text{is } \mathcal{F} \otimes \mathcal{B}_\mathrm{T} \otimes \mathcal{B}_\mathrm{X}\text{-measurable.}$$

($ii$) for fixed $\omega \in \Omega$ and $t \geqslant 0$, the mapping

$$\begin{array}{ccc} \mathrm{X} & \longrightarrow & \mathcal{M}_d(\mathbf{R}) \\ x & \longmapsto & \mathcal{A}^t_{\omega,x} \end{array} \quad \text{is continuous.}$$

($iii$) (cocycle property): $\quad \forall\, x \in \mathrm{X}, \quad \forall\, \omega \in \Omega, \quad \forall\, s, t \in \mathrm{T}$

$$\mathcal{A}^{t+s}_{\omega,x} \,=\, \mathcal{A}^t_{\Theta^s(\omega,x)}\, \mathcal{A}^s_{\omega,x}$$

and $\mathcal{A}^0_{\omega,x} \,=\, \mathrm{I}_d$.

We call the **generator** of $\mathcal{A}$ the mapping: $A \,=\, \mathcal{A}^1_{(\cdot,\cdot)} \,:\, \Omega \times \mathrm{X} \;\longrightarrow\; \mathcal{M}_d(\mathbf{R})$.

**Example 1.** Let X be a Euclidean space and assume that $\mathcal{T}$ is $\mathcal{C}^1$ on X. Then the following

$$\mathcal{A}^t_{\omega,x} \;:=\; D_x \mathcal{T}^t_\omega,$$

is a linear cocycle and is often referred to as the *derivative cocycle* for $\mathcal{T}$. Indeed, the cocycle property of $\mathcal{A}$ is a manifestation of the Chain Rule from standard calculus and the cocycle property of $\mathcal{T}$.

**Definition 23.** We say that $\mathcal{A}$ is **invertible** if:

$$\forall\, \omega \in \Omega, \quad \forall\, t \in \mathrm{T}, \quad \forall\, x \in \mathrm{X} \;:\quad \mathcal{A}^t_{\omega,x} \in \mathrm{GL}_d(\mathbf{R}).$$

For $a \in \mathbf{R}$, we denote $a^+ \;:=\; \max(0, a)$.

**Definition 24.** Assume that, in addition, $\mathcal{T}$ has independent increments and admits a stationary measure $\mu$. We then say that the invertible cocycle $\mathcal{A}$ is **integrable** if:

$$\mathbb{E}\left(\int \alpha^+(\cdot, x)\, \mathrm{d}\mu(x)\right), \quad \mathbb{E}\left(\int \alpha^-(\cdot, x)\, \mathrm{d}\mu(x)\right) \,<\, +\infty,$$

where: $\quad \alpha^+(\omega, x) \;:=\; \sup_{0 \leqslant t \leqslant 1} \log^+ \left\|\mathcal{A}^t_{\omega,x}\right\| \qquad \text{and} \qquad \alpha^-(\omega, x) \;:=\; \sup_{0 \leqslant t \leqslant 1} \log^+ \left\|(\mathcal{A}^t_{\omega,x})^{-1}\right\|.$

### 3.2. **The Subadditive Ergodic Theorem**

As a final preparation for the MET, we prove a theorem of Furstenberg and Kesten [16]. We hence need several versions of the latter. We first introduce a generalization of subadditivity to stochastic processes. Let $(\mathrm{E}, \mathcal{E}, \nu, (\vartheta^t)_{t \in \mathrm{T}})$ be a metric DS.

**Definition 25** (Subadditive Stochastic Process)**.** A measurable stochastic process $(g_t(\omega))_{t\in \mathrm{T}}$ with values in $\mathbf{R}\cup\{-\infty\}$ is called **subadditive** over $(\mathrm{E},\mathcal{E},\nu,(\vartheta^t))$ if

$$g_{s+t}(\omega) \leqslant g_s(\omega) + g_t(\vartheta^s\omega)$$

holds identically. We call $g$ **superadditive** if $-g$ is subadditive.

**Example 2.** Let $g : \mathrm{E} \longrightarrow \mathbf{R}$ be a measurable function such that $t \longmapsto g(\vartheta^t\omega)$ is locally integrable. Then

$$S_t(\omega) := \int_0^t g(\vartheta^s\omega)\,\mathrm{d}s,$$

is are sub- as well as superadditive.

**Theorem 12** (Kingman's Subadditive Ergodic Theorem, [30])**.** *Let $(f_n)$ be a subadditive sequence of random variables over $(\mathrm{E},\mathcal{E},\nu,(\vartheta^n)_{n\in\mathbf{N}})$. Assume $f_1^+ \in \mathrm{L}^1(\mathrm{E},\mathcal{E},\nu)$ and put*

$$m := \inf\left\{\frac{1}{n}\mathbb{E}(f_n),\ n\in\mathbf{N}\right\}.$$

*Then:*

*(a) there exists:*

- *a forward $\vartheta$-invariant set $\mathrm{U}\in\mathcal{E}$ of full $\nu$-measure,*
- *a measurable function $\gamma : \mathrm{E}\longrightarrow \mathbf{R}\cup\{-\infty\}$ with $\gamma^+\in\mathrm{L}^1$ such that on $\mathrm{U}$:*
  - $\forall\,\omega\in\mathrm{U}\ :\quad \lim\limits_{n+\infty}\left(\dfrac{1}{n}f_n(\omega)\right) = \gamma(\omega),$
  - $\forall\,k\in\mathbf{N}\ :\quad \gamma(\vartheta^k\omega) = \gamma(\omega),$

  *and $\gamma(\omega) = m$ in the ergodic case.*

*Further:*

$$\lim_{n+\infty}\frac{1}{n}\mathbb{E}(f_n) = m = \mathbb{E}(\gamma) \in \mathbf{R}\cup\{-\infty\}.$$

*(b) In addition, if:*

$$m > -\infty \qquad \text{and} \qquad \forall\, n\in\mathbf{N}\ :\ f_n\in\mathrm{L}^1,$$

*then assertion $(a)$ holds with $\gamma\in\mathrm{L}^1$, and convergence in $(i)$ also holds in $\mathrm{L}^1$.*

*Proof.* We omit the proof as it is easily available in the literature, see e.g. Steele [30]. □

*Remark.* We will use this fundamental theorem with:

$$\mathrm{E} = \Omega\times\mathrm{X}, \qquad \mathcal{E} = \mathcal{F}\otimes\mathcal{B}_{\mathrm{X}}, \qquad \vartheta = \Theta \qquad \text{and} \qquad \nu = (\mathbb{P}\times\mu).$$

### 3.3. **The Furstenberg-Kesten Theorem**

In 1960, Furstenberg and Kesten proved a Strong Law of Large Numbers for random matrices, with some restrictive positivity conditions on the entries, and without computing the limits explicitly. In 1963, Furstenberg provided a formula for almost sure limits for normalized products of random matrices in terms of a stationary measure. Despite a considerable amount of further work, the invariant (stationary) measures, and hence the limits in the strong law, had not been found explicitly.

**Theorem 13** (Furstenberg-Kesten [16])**.** *Assume that, in addition, $\mathcal{T}$ has independent increments and admits a stationary measure $\mu$. Let $\mathcal{A}$ be a linear invertible and integrable cocycle. Then:*

(*i*) *for each* $k \in [\![1,d]\!]$ *the sequence*

$$\left(f_t^{(k)}(\omega,x) \;:=\; \log\left\|\wedge^k \mathcal{A}^t_{\omega,x}\right\|\right)_{t\geqslant 0},$$

*is subadditive and* $f_1^{(k)+} \in \mathrm{L}^1(\Omega\times \mathrm{X},\, \mathcal{F}\otimes\mathcal{B}_{\mathrm{X}},\, \mathbb{P}\times\mu)$.

(*ii*) *there exists:*

- *a* $(\Theta^t)$*-invariant set* $\mathrm{U}\in\mathcal{F}\otimes\mathcal{B}_{\mathrm{X}}$ *of full* $(\mathbb{P}\times\mu)$*-measure,*
- *measurable functions* $\gamma^{(k)}:\Omega\times\mathrm{X}\longrightarrow\mathbf{R}$ *with* $\gamma^{(k)^+}\in\mathrm{L}^1$ *such that on* U :
  - $\gamma^{(k)}(\omega,x) \;=\; \lim\limits_{t+\infty}\dfrac{1}{t}\log\left\|\wedge^k\mathcal{A}^t_{\omega,x}\right\|,$
  - $\gamma^{(k)}(\Theta^t(\omega,x)) \;=\; \gamma^{(k)}(\omega,x),$
  - $\gamma^{(k+m)}(\omega,x) \;\leqslant\; \gamma^{(k)}(\omega,x)+\gamma^{(m)}(\omega,x),$

  *and:* $\gamma^{(k)}(\omega,x) \;=\; \mathbb{E}(\gamma^{(k)})$ *in the ergodic case.*

*Further,*

$$\lim_{t+\infty}\frac{1}{t}\mathbb{E}\left(\log\left\|\wedge^k\mathcal{A}^t_{(\cdot,\cdot)}\right\|\right) \;=\; \mathbb{E}(\gamma^{(k)}) \;=\; \inf\left\{\frac{1}{t}\mathbb{E}\left(\log\left\|\wedge^k\mathcal{A}^t_{(\cdot,\cdot)}\right\|\right),\; t\;\geqslant\;0\right\},$$

*is finite and converge in* $\mathrm{L}^1$.

(*iii*) *The measurable functions* $\chi_k$ *successively defined by*

$$\forall\, k\in[\![1,d]\!]\;:\quad \chi_1(\omega,x)+\ldots+\chi_k(\omega,x) \;:=\; \gamma^{(k)}(\omega,x),$$

*have the following properties on* U :

- $\chi_k(\omega,x) \;=\; \lim\limits_{t+\infty}\dfrac{1}{t}\log\sigma_k(\mathcal{A}^t_{\omega,x}),$
- $\chi_k(\Theta^t(\omega,x)) \;=\; \chi_k(\omega,x),$
- $\chi_1(\omega,x)\;\geqslant\;\chi_2(\omega,x)\;\geqslant\;\ldots\;\geqslant\;\chi_d(\omega,x),$

*and:* $\chi_k(\omega,x) \;=\; \mathbb{E}(\chi_k)$ *in the ergodic case. Further:*

$$\lim_{t+\infty}\frac{1}{t}\mathbb{E}\left(\log\sigma_k(\mathcal{A}^t_{(\cdot,\cdot)})\right) \;=\; \mathbb{E}(\chi_k).$$

*Remark* (Furstenberg-Kesten Theorem for Other Norms). We work with the standard scalar product in $\mathbf{R}^d$ to define "orthogonal", "singular value decomposition", the corresponding operator matrix norms in $\wedge^k\mathbf{R}^d$, etc. However, as all norms in $\mathcal{M}_d(\mathbf{R})$ are equivalent, the integrability condition is satisfied or not for all norms simultaneously, and all limits are independent of the norm chosen.

However, the representation

$$\mathbb{E}(\gamma^{(k)}) \;=\; \inf_{n\in\mathbf{N}}\frac{1}{n}\mathbb{E}\left(\log\left\|\wedge^k\mathcal{A}^n_{(\cdot,\cdot)}\right\|\right)$$

is only valid for so-called matrix norms (*i.e.*, norms on $\mathcal{M}_d(\mathbf{R})$ with the additional property $\|AB\|\;\leqslant\;\|A\|\|B\|$).

**Lemma 14** (One-Sided Discrete Time)**.** *Let* $\mathcal{A}$ *be a linear cocycle and consider* $A$ *its generator. Then the following statements hold:*

*(A) Non-invertible case: if*

$$\log^+ \left\| \mathcal{A}^1_{(\cdot,\cdot)} \right\| \in \mathrm{L}^1(\Omega \times \mathrm{X},\, \mathcal{F} \otimes \mathcal{B}_{\mathrm{X}},\, \mathbb{P} \times \mu),$$

*then there exists a $\Theta$-invariant set $\mathrm{U} \in \mathcal{F} \otimes \mathcal{B}_{\mathrm{X}}$ of full $(\mathbb{P} \times \mu)$-measure on which the $(\omega, x)$-wise statements of the previous theorem hold with $t$ and $\Theta^t$ replaced with $n$ and $\Theta$, and except that all $\gamma^{(k)}$ (hence all $\chi_k$) aren't especially finite.*

*(B) Invertible case: if $\mathcal{A}^1_{(\cdot,\cdot)} : \Omega \longrightarrow \mathrm{GL}_d(\mathbf{R})$ and*

$$\log^+ \|\mathcal{A}^1_{(\cdot,\cdot)}\| \in \mathrm{L}^1 \qquad \text{and} \qquad \log^+ \|(\mathcal{A}^1_{(\cdot,\cdot)})^{-1}\| \in \mathrm{L}^1,$$

*then there exists a $\Theta$-invariant set $\mathrm{U} \in \mathcal{F} \otimes \mathcal{B}_{\mathrm{X}}$ of full $(\mathbb{P} \times \mu)$-measure on which the $(\omega, x)$-wise statements of the previous theorem hold with $t$ and $\Theta^t$ replaced with $n$ and $\Theta$.*

*Proof of the lemma.*

(A) $(i)$ The cocycle property for $\wedge^k \mathcal{A}^n_{\omega,x}$ and Lemma 56 $(vi)$ imply that

$$f^{(k)}_n(\omega, x) \;:=\; \log \left\| \wedge^k \mathcal{A}^n_{\omega,x} \right\|,$$

is subadditive. Furthermore:

$$f^{(1)+}_1 \in \mathrm{L}^1 \qquad \text{and} \qquad \left\| \wedge^k A \right\| \leqslant \|A\|^k \qquad \Longrightarrow \qquad f^{(k)+}_1 \in \mathrm{L}^1.$$

$(ii)$ This is an immediate consequence of Kingman's Subadditive Ergodic Theorem applied for each $k$. In particular, the forward invariant set U can be taken to be the intersection of the $d$ forward invariant sets $\mathrm{U}_k$ for $k \in [\![1, d]\!]$. Moreover:

$$\left\| \wedge^{k+\ell} \mathcal{A} \right\| \;\leqslant\; \left\| \wedge^k \mathcal{A} \right\| \left\| \wedge^\ell \mathcal{A} \right\| \quad \Longrightarrow \quad \gamma^{(k+\ell)}(\omega, x) \leqslant \gamma^{(k)}(\omega, x) + \gamma^{(\ell)}(\omega, x).$$

In particular, if $\gamma^{(k)}(\omega, x) = -\infty$ for some $k$, then $\gamma^{(\ell)}(\omega, x) = -\infty$ for all $\ell \geqslant k$.

$(iii)$ By Lemma 57 $(iii)$ for $k \in [\![1, d]\!]$ we have that:

$$\frac{1}{n} \log \left\| \wedge^k \mathcal{A}^n_{\omega,x} \right\| \;=\; \frac{1}{n} \sum_{i=1}^{k} \log \sigma_i(\mathcal{A}^n_{\omega,x}).$$

We proceed successively as follows: choose $\omega \in \mathrm{U}$.

- For $k = 1$:

$$\gamma^{(1)}(\omega, x) \;=\; \lim_{n+\infty} \frac{1}{n} \log \sigma_1(\mathcal{A}^n_{\omega,x}) \;=:\; \chi_1(\omega, x) \;\text{ exists.}$$

- For $k = 2$: if $\gamma^{(1)}(\omega, x) = -\infty$, we are finished. If $\gamma^{(1)}(\omega, x) > -\infty$:

$$\gamma^{(2)}(\omega, x) - \gamma^{(1)}(\omega, x) \;=\; \lim_{n+\infty} \frac{1}{n} \log \sigma_2(\mathcal{A}^n_{\omega,x}) \;=:\; \chi_2(\omega, x) \;\text{ exists,}$$

and so on (for $3 \leqslant k \leqslant d-1$).

- For $k = d$: if finally $\gamma^{(d-1)}(\omega, x) > -\infty$,

$$\gamma^{(d)}(\omega, x) - \gamma^{(d-1)}(\omega, x) \;=\; \lim_{n+\infty} \frac{1}{n} \log \sigma_d(\mathcal{A}^n_{\omega,x}) \;=:\; \chi_d(\omega, x) \;\text{ exists.}$$

By this construction,

$$\gamma^{(k)}(\omega,x) \,=\, -\infty \qquad \Longleftrightarrow \qquad \chi_k(\omega,x) \,=\, -\infty.$$

The inequalities $\chi_1(\omega,x) \,\geqslant\, \ldots \,\geqslant\, \chi_d(\omega,x)$ on U follow from $\sigma_1 \,\geqslant\, \ldots \,\geqslant\, \sigma_d$.

For the expectations, the limit

$$\mathbb{E}(\gamma^{(1)}) \,=\, \lim_{n+\infty} \frac{1}{n}\mathbb{E}\left(\log \sigma_1(\mathcal{A}^n_{(\cdot,\cdot)})\right) \,=\, \mathbb{E}(\chi_1) \text{ exists.}$$

If $\mathbb{E}(\gamma^{(1)}) \,=\, -\infty$ we are done as all other expectations $\mathbb{E}(\gamma^{(k)})$ are also equal to $-\infty$. If not, the limit

$$\mathbb{E}(\gamma^{(2)}) - \mathbb{E}(\gamma^{(1)}) \,=\, \lim_{n+\infty} \frac{1}{n}\mathbb{E}\left(\log \sigma_2(\mathcal{A}^n_{(\cdot,\cdot)})\right) \,=\, \mathbb{E}(\chi_2)$$

exists, and so on.

(B) We want to apply part $(b)$ of Kingman's Subadditive Ergodic Theorem, for which we have to verify that for all $n$ and $k$ :

$$f_n^{(k)} \in \mathrm{L}^1 \qquad \text{and} \qquad \mathbb{E}(\gamma^{(k)}) \,>\, -\infty.$$

(1) From $1 \,=\, \left\|AA^{-1}\right\| \,\leqslant\, \|A\| \left\|A^{-1}\right\|$ it follows that

$$-\log^+ \left\|A^{-1}\right\| \,\leqslant\, \log \|A\| \,\leqslant\, \log^+ \|A\|$$

and

$$\left|\log \|A\|\right| \,\leqslant\, \log^+ \|A\| + \log^+ \left\|A^{-1}\right\|.$$

Similarly,

$$\left|\log \left\|\wedge^k A\right\|\right| \,\leqslant\, k\left(\log^+ \|A\| + \log^+ \left\|A^{-1}\right\|\right).$$

Therefore, since $\log^+ \left\|A^{\pm 1}\right\| \in \mathrm{L}^1$, we have that $f_1^{(k)} \in \mathrm{L}^1$.

(2) Since $\log \left\|\wedge^k A\right\| \,\leqslant\, k \log^+ \|A\|$ and:

$$\left\|\wedge^k \mathcal{A}^{n+1}_{\omega,x}\right\| \,\leqslant\, \left\|\wedge^k \mathcal{A}^n_{\Theta(\omega,x)}\right\| \left\|\wedge^k A(\omega,x)\right\| \quad \text{and} \quad \left\|\wedge^k \mathcal{A}^n_{\Theta(\omega,x)}\right\| \,\leqslant\, \left\|\wedge^k \mathcal{A}^{n+1}_{\Theta(\omega,x)}\right\| \left\|\wedge^k A(\omega,x)^{-1}\right\|,$$

we have:

$$(f_n^{(k)} \circ \Theta) - k \log^+ \left\|A^{-1}\right\| \;\leqslant\; f_{n+1}^{(k)} \;\leqslant\; (f_n^{(k)} \circ \Theta) + k \log^+ \|A\|. \qquad (*)$$

Since $\mathbb{P} \times \mu$ is $\Theta$-invariant and since $\log^+ \left\|(A)^{\pm 1}\right\| \in \mathrm{L}^1$, we have by (1) that $f_n^{(k)} \in \mathrm{L}^1$.

(3) By (2) we have:

$$\begin{aligned} \left|\mathbb{E}(\gamma^{(k)})\right| \;&=\; \lim_{n+\infty} \frac{1}{n} \left|\mathbb{E}\left(\log \left\|\wedge^k \mathcal{A}^n_{\omega,x}\right\|\right)\right| \\ &\leqslant\; k \lim_{n+\infty} \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E}\left|\log \left\|A(\Theta^i(\omega,x))\right\|\right| \,=\, k\mathbb{E}\left|\log \|A\|\right| \,<\, \infty. \end{aligned}$$

(4) Dividing equation $(*)$ by $n$ we see that

$$\frac{f_{n+1}^{(k)}(\omega,x)}{n} \;\xrightarrow[n+\infty]{}\; \gamma^{(k)}(\omega,x) \qquad \Longleftrightarrow \qquad \frac{f_n^{(k)}(\Theta(\omega,x))}{n} \;\xrightarrow[n+\infty]{}\; \gamma^{(k)}(\Theta(\omega,x))$$

and in that case $\gamma^{(k)}(\Theta(\omega,x)) \,=\, \gamma^{(k)}(\omega,x)$. Hence the set U on which the limit exists and is finite for each $k$ is invariant whenever $A(\omega,x) \in \mathrm{GL}_d(\mathbf{R})$ for all $\omega \in \Omega$ and $x \in \mathrm{X}$. On this set, the limits are invariant. We know from Lemma 14 $(A)(i)$ that U has full measure.

□

*Proof of Furstenberg-Kesten Theorem.* Since $\mathcal{A}$ is integrable, then $\log^+ \left\|(\mathcal{A}^1_{\omega,x})^{\pm 1}\right\| \in \mathrm{L}^1$ and part $(B)$ of Lemma 14 applies. Hence, let $\mathrm{U}_1$ be the $\Theta^1$ invariant set of full measure of those $\omega$'s for which the limits $\gamma^{(k)}(\omega, x)$ exist and are finite. These limits are $\Theta^1$-invariant on $\mathrm{U}_1$.

(1) By the continuity of $\mathcal{A}$ with respect to $t$, we have that $\alpha^{\pm}$ is measurable. Since $\alpha^{\pm} \in \mathrm{L}^1$ and the ergodic theorem assure that the $\Theta^1$-invariant set $\mathrm{U}_2$ on which

$$\lim_{n+\infty} \frac{1}{n} \alpha^{\pm}(\Theta^n(\omega, x)) \,=\, 0,$$

has full measure. Hence $\mathrm{U} \,:=\, \mathrm{U}_1 \cap \mathrm{U}_2$ is $\Theta^1$-invariant and has full measure.

(2) Let's now prove that $\mathrm{U}_2$ is $\Theta^t$-invariant. For all $t \in [0,1]$ :

$$\begin{aligned}
\alpha^+(\Theta^t(\omega,x)) &= \sup_{0\leqslant s\leqslant 1} \log^+ \left\|(\mathcal{A}^{s+t}_{\omega,x})(\mathcal{A}^t_{\omega,x})^{-1}\right\| \\
&\leqslant \sup_{0\leqslant s\leqslant 2} \log^+ \left\|\mathcal{A}^s_{\omega,x}\right\| + \alpha^-(\omega,x) \\
&\leqslant \alpha^+(\omega,x) + \sup_{0\leqslant s\leqslant 1} \log^+ \left\|\mathcal{A}^s_{\Theta^1(\omega,x)}\mathcal{A}^1_{\omega,x}\right\| + \alpha^-(\omega,x) \\
&\leqslant 2\alpha^+(\omega,x) + \alpha^+(\Theta^1(\omega,x)) + \alpha^-(\omega,x)
\end{aligned}$$

similarly for $\alpha^-$. Hence $\mathrm{U}_2$ is forward invariant.

(3) For the converse: since for $s,t \in [0,1]$

$$\mathcal{A}^{s+1-t}_{\Theta^t(\omega,x)} \,=\, \mathcal{A}^s_{\Theta^1(\omega,x)} \mathcal{A}^{1-t}_{\Theta^t(\omega,x)},$$

then we have

$$\alpha^+(\Theta^1(\omega,x)) \;\leqslant\; 2\alpha^+(\Theta^t(\omega,x)) + \alpha^+(\Theta^1\Theta^t(\omega,x)) + \alpha^-(\Theta^t(\omega,x)),$$

similarly for $\alpha^-$. Hence $\Theta^t(\omega,x) \in \mathrm{U}_2$ implies $\Theta^1(\omega,x) \in \mathrm{U}_2$ which holds if and only if $\omega \in \mathrm{U}_2$.

(4) We now prove that for each $\omega \in \mathrm{U}$ the limit

$$\lim_{t+\infty} \frac{1}{t} \log \left\|\wedge^k \mathcal{A}^t_{\omega,x}\right\| \;\text{ exists and is equal to } \gamma^{(k)}(\omega,x).$$

The cocycle property gives for $t \geqslant 0$ and $n \,=\, [t]$,

$$\wedge^k \mathcal{A}^t_{\omega,x} \,=\, (\wedge^k \mathcal{A}^{t-n}_{\Theta^n(\omega,x)})(\wedge^k \mathcal{A}^n_{\omega,x}),$$

whence by the invertibility of $\mathcal{A}$

$$\wedge^k \mathcal{A}^n_{\omega,x} \,=\, \wedge^k (\mathcal{A}^{t-n}_{\Theta^n(\omega,x)})^{-1} (\wedge^k \mathcal{A}^t_{\omega,x}).$$

Hence:

$$\begin{aligned}
\log \left\|\wedge^k \mathcal{A}^n_{\omega,x}\right\| - k\alpha^-(\Theta^n(\omega,x)) &\leqslant \log \left\|\wedge^k \mathcal{A}^t_{\omega,x}\right\| &(1)\\
&\leqslant \log \left\|\wedge^k \mathcal{A}^n_{\omega,x}\right\| + k\alpha^+(\Theta^n(\omega,x)) &(2)
\end{aligned}$$

We conclude from (1) that for $\omega \in \mathrm{U}$ the limits exist and are equal to $\gamma^{(k)}(\omega,x)$

(5) We finally prove that U is $\Theta^t$-invariant:

$$(\Theta^t)^{-1}\mathrm{U} \,=\, \mathrm{U} \qquad \text{and} \qquad \forall\, \omega \in \mathrm{U}, \quad \forall\, t \,\geqslant\, 0 \;\; : \;\; \gamma^{(k)}(\Theta^t(\omega,x)) \,=\, \gamma^{(k)}(\omega,x).$$

Consider the case $k = 1$ only. Let $n \,=\, [t]$, and assume that $s \,\geqslant\, n+1-t$. By the cocycle property,

$$\begin{aligned} \log \left\| \mathcal{A}^{t+s-(n+1)}_{\Theta^{n+1}(\omega,x)} \right\| - \log \left\| (\mathcal{A}^{n+1-t}_{\Theta^t(\omega,x)})^{-1} \right\| \;&\leqslant\; \log \left\| \mathcal{A}^{s}_{\Theta^t(\omega,x)} \right\| \qquad (3) \\ &\leqslant\; \log \left\| \mathcal{A}^{t+s-(n+1)}_{\Theta^{n+1}(\omega,x)} \right\| + \log \left\| \mathcal{A}^{n+1-t}_{\Theta^t(\omega,x)} \right\|. \qquad (4) \end{aligned}$$

Hence, when this inequality is divided by $s$ and letting $[s \to +\infty]$, we have that:

$$\begin{cases} \lim\limits_{s+\infty} \dfrac{1}{s} \left( \log \left\| \mathcal{A}^{t+s-(n+1)}_{\Theta^{n+1}(\omega,x)} \right\| - \log \left\| (\mathcal{A}^{n+1-t}_{\Theta^t(\omega,x)})^{-1} \right\| \right) \;=\; \gamma^{(1)}(\Theta^{n+1}(\omega,x)), \\ \\ \lim\limits_{s+\infty} \dfrac{1}{s} \left( \log \left\| \mathcal{A}^{s}_{\Theta^t(\omega,x)} \right\| \right) \qquad\qquad\qquad = \qquad \gamma^{(1)}(\omega,x), \end{cases}$$

thus:

$$\gamma^{(1)}(\Theta^{n+1}(\omega,x)) \,=\, \gamma^{(1)}(\omega,x)$$

Consequently, $\Theta^t(\omega,x) \in \mathrm{U}$, *i.e.*, U is forward invariant. Moreover,

$$\lim_{s+\infty} \frac{1}{s} \log \left\| \mathcal{A}^{s}_{\Theta^t(\omega,x)} \right\| \,=\, \lim_{n+\infty} \frac{1}{n} \log \left\| \mathcal{A}^{n}_{\Theta^t(\omega,x)} \right\| \,=\, \gamma^{(1)}(\Theta^t(\omega,x)) \,=\, \gamma^{(1)}(\omega,x)$$

Rearranging the string of inequalities (3), we can also conclude that if $\Theta^t(\omega,x) \in \mathrm{U}$, then $\omega \in \mathrm{U}$, *i.e.*, U is invariant.

□

**Corollary 15** (Lyapunov Index of Determinant)**.** *We have*

$$\lim_{n+\infty} \frac{1}{n} \log |\det \mathcal{A}^{n}_{(\cdot,\cdot)}| \;=\; \gamma^{(d)} \;=\; \sum_{k=1}^{d} \chi_k \;=\; \mathbb{E}\left(\log |\det A(\cdot)| \,|\, \mathcal{I}\right) \quad \mathbb{P}\text{-a.s.}$$

*where $\mathcal{I} \subset \mathcal{F}$ is the $\sigma$-algebra of $\theta$-invariant sets, and*

$$\mathbb{E}(\gamma^{(d)}) \,=\, \sum_{k=1}^{d} \mathbb{E}(\chi_k) \,=\, \mathbb{E}(\log |\det A|) \;\in\; \mathbf{R} \cup \{-\infty\}$$

*where the expectations are finite when* $\log^+ \left\| A^{\pm 1} \right\| \in \mathrm{L}^1$.

*Proof.* By Proposition 57 (*iii*):

$$\left\| \wedge^d \mathcal{A}^{n}_{\omega,x} \right\| \,=\, |\det \mathcal{A}^{n}_{\omega,x}| \,=\, \prod_{i=0}^{n-1} \left| \det A(\Theta^i(\omega,x)) \right|.$$

We know that $\frac{1}{n}\log(\cdot)$ of the left-hand side converges on U to

$$\gamma^{(d)} \,=\, \sum_{k=1}^{d} \chi_k \,=\, \lim_{n+\infty} \frac{1}{n} \log \left\| \wedge^d \mathcal{A}^{n}_{\omega,x} \right\|$$

The (generalized) ergodic theorem applies to the right-hand side since $\log^+ |\det A| = f_1^{(d)+} \in \mathrm{L}^1$ and yields

$$\lim_{n+\infty} \frac{1}{n} \sum_{i=0}^{n-1} \log \left| \det A(\Theta^i(\omega,x)) \right| \,=\, \mathbb{E}(\log |\det A(\cdot)| \,|\, \mathcal{I}) \in \mathbf{R} \cup \{-\infty\} \quad \mathbb{P}\text{-a.s.}$$

□

**Definition of the Lyapunov spectrum**

We can now define the main tool for studying Lagrangian chaos. Denote by $\lambda_1(\omega,x) > \ldots > \lambda_{p(\omega,x)}(\omega,x)$ the different numbers in the sequence $\chi_1(\omega,x) \geqslant \ldots \geqslant \chi_d(\omega,x)$ and denote by $d_i(\omega,x)$ the frequency of appearance of $\lambda_i(\omega,x)$ in this sequence.

On the corresponding forward invariant or invariant set U of full measure the functions

$$\begin{array}{rccc} p: & \mathrm{U} & \longrightarrow & [\![1,d]\!] \\ d_i: & \{(\omega,x)\in \mathrm{U} : p(\omega,x) \geqslant i\} & \longrightarrow & [\![1,d]\!] \\ \lambda_i: & \{(\omega,x)\in \mathrm{U} : p(\omega,x) \geqslant i\} & \longrightarrow & \mathbf{R}\cup\{-\infty\}, \end{array}$$

are invariant, and constant in the ergodic case.

**Definition 26** (Lyapunov Spectrum)**.** We then call:

- the **Lyapunov exponents**: the functions $\lambda_i(\cdot)$,
- their **multiplicities**: the functions $d_i(\cdot)$,
- the **Lyapunov spectrum**: the set

$$\mathscr{S}(\theta,\mathcal{A}) \;:=\; \{(\lambda_i(\cdot),d_i(\cdot)),\; i\in[\![1,p(\cdot)]\!]\}\,.$$

*Remark.* By remark on Furstenberg-Kesten for other norms, the Lyapunov spectrum is independent of the choice of the norm in the space $\mathcal{M}_d(\mathbf{R})$. If $\Theta$ is ergodic, we will agree that the Lyapunov spectrum consists of the corresponding set of constants.

## 3.4. The Multiplicative Ergodic Theorem

**Statement of the theorem**

**Theorem 16** (Multiplicative Ergodic Theorem)**.** *Let:*

(*i*) $\mathcal{T}$ *be a strongly continuous RDS with independent increments, and which admits:*

(*ii*) $\mu$ *be a stationary measure for* $\mathcal{T}$*,*

(*iii*) $\mathcal{A}$ *be an invertible and integrable linear cocycle.*

*Then, there exist:*

- *a* $(\Theta^t)$*-invariant set* $\mathrm{U}\subset\Omega\times\mathrm{X}$ *of full* $(\mathbb{P}\times\mu)$*-measure,*
- *a flag of subspaces for each* $(\omega,x)\in\mathrm{U}$ *:*

$$\{0\} \;=:\; \mathrm{F}_{p(\omega,x)+1} \;\subset\; \mathrm{F}_{p(\omega,x)}(\omega,x) \;\subset\; \cdots \;\subset\; \mathrm{F}_2(\omega,x) \;\subset\; \mathrm{F}_1 \;:=\; \mathbf{R}^d,$$

*where* $\dim \mathrm{F}_i(\omega,x)\in[\![1,d]\!]$ *and for which:*

$$\forall\, v\in\mathbf{R}^d\backslash\{0\} \;:\quad \lambda(\omega,x,v) \;:=\; \lim_{t+\infty}\frac{1}{t}\log\|\mathcal{A}^t_{\omega,x}v\| \;\text{ exists as a limit}$$

*and*

$$\lambda(\omega,x,v) \;=\; \lambda_i(\omega,x) \qquad\Longleftrightarrow\qquad v\in\mathrm{F}_i(\omega,x)\backslash\mathrm{F}_{i+1}(\omega,x).$$

*Equivalently:*

$$\mathrm{F}_i(\omega,x) \;=\; \left\{v\in\mathbf{R}^d : \lambda(\omega,x,v) \;\leqslant\; \lambda_i(\omega,x)\right\}.$$

*If* $\mu$ *is ergodic, then the function* $p(\cdot)$ *is constant on* U*, and for* $i\in[\![1,d]\!]$ *the functions* $\lambda_i(\cdot)$ *and* $d_i(\cdot)$ *are constant on* $\{\omega\in\mathrm{U} : p(\omega,x)\geqslant i\}$*. Further:*

$$\forall\, i\in[\![1,r]\!],\quad \forall\, v\in\mathrm{F}_i(\omega,x)\backslash\mathrm{F}_{i+1}(\omega,x) \;:\quad \lim_{t+\infty}\frac{1}{t}\log\left\|\mathcal{A}^t_{\omega,x}v\right\| \;=\; \lambda_i.$$

**Proof of the MET**

We prove the MET for Euclidean spaces. First, we prove the theorem for discrete time. This following lemma is fundamental and we will obtain these stochastic results by exploiting a matrix algebra result.

**Lemma 17** (MET for One-Sided Discrete Time)**.** *Let $\mathcal{A}$ be a linear cocycle and consider $A$ its generator. Then the following statements hold:*

*(A) Non-invertible case* $\mathrm{T} = \mathbf{N}$ *: if*

$$\log^+ \left\| \mathcal{A}^1_{(\cdot,\cdot)} \right\| \in \mathrm{L}^1,$$

*then there exists a forward $\Theta$-invariant set $\mathrm{U} \in \mathcal{F} \otimes \mathcal{B}_{\mathrm{X}}$ of full $(\mathbb{P} \times \mu)$-measure such that for each $(\omega, x) \in \mathrm{U}$:*

*(i) the limit:*

$$\lim_{n+\infty} \left( (\mathcal{A}^n_{\omega,x})^\top \mathcal{A}^n_{\omega,x} \right)^{\frac{1}{2n}} =: \Lambda_{\omega,x} \geqslant 0 \text{ exists.}$$

*(ii) let $e^{\lambda_{p(\omega,x)}(\omega,x)} < \ldots < e^{\lambda_1(\omega,x)}$ be the different eigenvalues of $\Lambda_{\omega,x}$ (possibly $\lambda_{p(\omega,x)}(\omega,x) = -\infty$) and let $\mathrm{E}_{p(\omega,x)}(\omega,x), \ldots, \mathrm{E}_1(\omega,x)$ be the corresponding eigenspaces with multiplicities $d_i(\omega,x) := \dim \mathrm{E}_i(\omega,x)$. Then for all $i \in \{1, \ldots, p(\omega,x)\}$ :*

$$\begin{aligned} p(\Theta(\omega,x)) &= p(\omega,x), \\ \lambda_i(\Theta(\omega,x)) &= \lambda_i(\omega,x), \\ d_i(\Theta(\omega,x)) &= d_i(\omega,x). \end{aligned}$$

*(iii) put $\mathrm{F}_{p(\omega,x)+1}(\omega,x) := \{0\}$ and for $i \in \{1, \ldots, p(\omega,x)\}$ :*

$$\mathrm{F}_i(\omega,x) := \mathrm{E}_{p(\omega,x)} \oplus \ldots \oplus \mathrm{E}_i(\omega,x),$$

*so that*

$$\mathrm{F}_{p(\omega,x)}(\omega,x) \subset \ldots \subset \mathrm{F}_i(\omega,x) \subset \ldots \subset \mathrm{F}_1(\omega,x) = \mathbf{R}^d,$$

*defines a filtration of $\mathbf{R}^d$. Then for each $v \in \mathbf{R}^d \backslash \{0\}$ the Lyapunov exponent*

$$\lambda(\omega,x,v) := \lim_{n+\infty} \frac{1}{n} \log \|\mathcal{A}^n_{\omega,x} v\|$$

*exists as a limit and*

$$\lambda(\omega,x,v) = \lambda_i(\omega,x) \qquad \Longleftrightarrow \qquad v \in \mathrm{F}_i(\omega,x) \backslash \mathrm{F}_{i+1}(\omega,x),$$

*equivalently*

$$\mathrm{F}_i(\omega,x) = \left\{ v \in \mathbf{R}^d : \lambda(\omega,x,v) \leqslant \lambda_i(\omega,x) \right\}.$$

*(iv) for all $v \in \mathbf{R}^d \backslash \{0\}$*

$$\lambda(\Theta(\omega,x), \mathcal{A}^1_{\omega,x} v) = \lambda(\omega,x,v),$$

*whence*

$$\forall\, i \in \{1, \ldots, p(\omega,x)\} : \quad A(\omega,x) \mathrm{F}_i(\omega,x) \subset \mathrm{F}_i(\Theta(\omega,x)).$$

*(v) if $(\Omega \times \mathrm{X}, \mathcal{F} \otimes \mathcal{B}_{\mathrm{X}}, \mathbb{P} \times \mu, (\Theta^n))$ is ergodic, then the function $p(\cdot)$ is constant on $\mathrm{U}$, and for $i \in [\![1, d]\!]$ the functions $\lambda_i(\cdot)$ and $d_i(\cdot)$ are constant on $\{(\omega,x) \in \mathrm{U} : p(\omega,x) \geqslant i\}$.*

*(B) Invertible case* $\mathrm{T} = \mathbf{N}$ *: if* $A : \Omega \times \mathrm{X} \longrightarrow \mathrm{GL}_d(\mathbf{R})$ *and:*

$$\log^+ \|A\|,\ \log^+ \|A^{-1}\| \in \mathrm{L}^1,$$

*then the set* U *of full measure on which* $(A)$ *holds (and on which in the ergodic case* $p(\cdot), \lambda_i(\cdot), d_i(\cdot)$ *are constant) can be chosen to be invariant. Further,* $\lambda_{p(\omega,x)}(\omega, x) > -\infty$ *on* U*, and*

$$\forall\, i \in \{1, \dots, p(\omega, x)\} \ : \quad A(\omega, x)\mathrm{F}_i(\omega, x) \,=\, \mathrm{F}_i(\Theta(\omega, x)).$$

The proof of this lemma follows in a few lines from the Furstenberg-Kesten theorem and the following (deterministic) matrix algebra result called "deterministic MET".

**Lemma 18** (Deterministic MET)**.** *Let* $(A_n)_{n\in\mathbf{N}}$ *be a sequence of* $\mathcal{M}_d(\mathbf{R})$ *which satisfies the following two conditions:*

$(i)$ $\limsup_{n+\infty} \left( \frac{1}{n} \log \|A_n\| \right) \leqslant 0,$

$(ii)$ *Putting* $\Phi_n := A_n \cdots A_1,$

$$\forall\, i \in [\![1, d]\!] \ : \quad \lim_{n+\infty} \left( \frac{1}{n} \log \left\| \wedge^i \Phi_n \right\| \right) \,=\, \gamma^{(i)} \,\in\, \mathbf{R} \cup \{-\infty\} \ \text{ exists.}$$

*Then:*

- *the limit* $\lim_{n+\infty} (\Phi_n^* \Phi_n)^{1/2n} \,=:\, \Lambda \,\geqslant\, 0$ *exists,*
- *the eigenvalues of* $\Lambda$ *coincide with* $(e^{\chi_i})$*, where the* $\chi_i$ *are successively defined by* $\chi_1 + \ldots +$ $\chi_i \,=\, \gamma^{(i)}$ *(put* $\chi_i = -\infty$ *if* $\gamma^{(i)} = -\infty$*) and satisfy*

$$\forall\, i \in [\![1, d]\!] \ : \quad \chi_i \,=\, \lim_{n+\infty} \frac{1}{n} \log \sigma_i(\Phi_n).$$

- *let* $e^{\lambda_p} < \ldots < e^{\lambda_1}$ *be the different eigenvalues of* $\Lambda$ *(possibly* $\lambda_p = -\infty$*) and let* $\mathrm{E}_p, \ldots, \mathrm{E}_1$ *be their corresponding eigenspaces with multiplicities* $d_i = \dim \mathrm{E}_i$*. Write* $\mathrm{F}_{p+1} := \{0\}$*,*

$$\mathrm{F}_i \,:=\, \bigoplus_{i=1}^{p} \mathrm{E}_i,$$

*so that*

$$\mathrm{F}_p \subset \ldots \subset \mathrm{F}_i \subset \ldots \subset \mathrm{F}_1 \,=\, \mathbf{R}^d,$$

*defines a filtration of* $\mathbf{R}^d$*. Then for each* $x \in \mathbf{R}^d \backslash \{0\}$ *the Lyapunov exponent*

$$\lambda(x) \,:=\, \lim_{n+\infty} \frac{1}{n} \log \|\Phi_n x\|$$

*exists as a limit and*

$$\lambda(x) \,=\, \lambda_i \qquad \Longleftrightarrow \qquad x \in \mathrm{F}_i \backslash \mathrm{F}_{i+1},$$

*equivalently*

$$\mathrm{F}_i \,=\, \left\{ x \in \mathbf{R}^d \,:\, \lambda(x) \,\leqslant\, \lambda_i \right\}.$$

*Framework of the proof.* This lemma is fully proven in [1]. Let us give the general idea of the proof for the case $d = 2$.

1. We first consider the singular value decomposition of $\Phi_n = V_n D_n O_n$, where the diagonal matrix is $D_n = \mathrm{Diag}(\sigma_1(\Phi_n), \ldots, \sigma_d(\Phi_n))$. By assumption

$$\lim_{n+\infty} D_n^{1/n} \,=\, \mathrm{Diag}(e^{\chi_1}, \ldots, e^{\chi_d}).$$

2. Denote by $\lambda_1 > \ldots > \lambda_p$ the different numbers among the $\chi_i$, $\mathrm{U}_i$ the eigenspace of $(\Phi_n^*\Phi_n)^{\frac{1}{2n}}$ corresponding to eigenvalues $\sigma_{k(i)}(\Phi_n)^{1/n}$ and let $P_i(n)$ be the corresponding orthogonal projection. We now show that (both the eigenvalues and) the eigenspaces of $(\Phi_n^*\Phi_n)^{\frac{1}{2n}}$ converge to limits ($e^{\lambda_i}$ and) $\mathrm{U}_i$.

3. Hence, as the convergence of eigenspaces is equivalent to the convergence of the corresponding orthogonal projections, we prove that:

$$P_i(n) \underset{n+\infty}{\longrightarrow} P_i \qquad \text{and} \qquad (\Phi_n^*\Phi_n)^{\frac{1}{2n}} \longrightarrow \Lambda = \sum_{i=1}^{d} e^{\lambda_i} P_i.$$

   Finally, the previous convergence has just the right speed to ensure the remainder of the lemma.

The general idea of this particular case allows us to deduce (with harder work) a proof for the general case. □

To derive the MET from this lemma, part of the following elementary lemma is needed.

**Lemma 19** (Growth of a Stationary Sequence)**.** *Let $(\mathrm{E}, \mathcal{E}, \nu, (\vartheta^n)_{n\in\mathbf{N}})$ be a metric DS. Suppose we are given a random variable $f : \mathrm{E} \longrightarrow \mathbf{R} \cup \{-\infty\}$.*

*(i) If $f^+ \in \mathrm{L}^1$, then the invariant set*

$$\mathrm{E}_1 := \left\{\omega : \limsup_{n+\infty} \left(\frac{1}{n} f(\vartheta^{n-1}\omega)\right) \leqslant 0\right\} \text{ has full measure.}$$

*(ii) Let $f : \Omega \longrightarrow \mathbf{R}$. Then the invariant set*

$$\mathrm{E}_2 := \left\{\omega : \liminf_{n+\infty} \left(\frac{1}{n} f(\vartheta^{n-1}\omega)\right) \leqslant 0 \leqslant \limsup_{n+\infty} \left(\frac{1}{n} f(\vartheta^{n-1}\omega)\right)\right\} \text{ has full measure.}$$

*If in addition $f^+ \in \mathrm{L}^1$, then the invariant set*

$$\mathrm{E}_3 := \left\{\omega : \limsup_{n+\infty} \left(\frac{1}{n} f(\vartheta^{n-1}\omega)\right) = 0\right\} \text{ has full measure.}$$

*If $f^- = \max(0, -f) \in \mathrm{L}^1$, then the invariant set*

$$\mathrm{E}_4 := \left\{\omega : \liminf_{n+\infty} \left(\frac{1}{n} f(\vartheta^{n-1}\omega) = 0\right)\right\} \text{ has full measure.}$$

*(iii) Let $f : \mathrm{E} \longrightarrow \mathbf{R}$. If $f \in \mathrm{L}^1$, then the invariant set*

$$\mathrm{E}_5 := \left\{\omega : \lim_{n+\infty} \left(\frac{1}{n} f(\vartheta^{n-1}\omega)\right) = 0\right\} \text{ has full measure.}$$

*It also works if $(f \circ \vartheta - f)^+ \in \mathrm{L}^1$, or if $(f \circ \vartheta - f)^- \in \mathrm{L}^1$, both implying $f \circ \vartheta - f \in \mathrm{L}^1$ and $\mathbb{E}(f \circ \vartheta - f) = 0$).*

*Proof of the growth of a stationary sequence.* The invariance of $\mathrm{E}_1, \mathrm{E}_2, \mathrm{E}_3$ and $\mathrm{E}_4$ follows from their definition.

*(i)* That $\mathrm{E}_1$ has full measure is a consequence of the Borel-Cantelli lemma since

$$\sum_{n=1}^{\infty} \nu\left(\frac{1}{n} f(\vartheta^{n-1}\cdot) > \varepsilon\right) = \sum_{n=1}^{\infty} \nu(f > \varepsilon n) = \sum_{n=1}^{\infty} \nu\left(f^+ > \varepsilon n\right) \leqslant \frac{1}{\varepsilon}\mathbb{E}(f^+) < \infty,$$

where we have used the elementary fact that for $f \geqslant 0$ and any $\varepsilon > 0$

$$\varepsilon \sum_{n=1}^{\infty} \nu(f > \varepsilon n) \leqslant \mathbb{E}(f) = \varepsilon \int_0^{\infty} \nu(f > \varepsilon t)\mathrm{d}t \leqslant \varepsilon \sum_{n=0}^{\infty} \nu(f > \varepsilon n)$$

(*ii*) If $f$ is finite-valued, then for any $\varepsilon > 0$

$$\lim_{n+\infty} \nu\left(\left|\frac{1}{n} f(\vartheta^{n-1}\omega)\right| > \varepsilon\right) = \lim_{n+\infty} \nu(|f(\omega)| > \varepsilon n) = 0,$$

whence $\frac{1}{n} f(\vartheta^{n-1}\cdot) \xrightarrow[n+\infty]{} 0$ in probability, from which the assertion follows. If $f^+ \in \mathrm{L}^1$ combine (*i*) with what we have just proved.

(*iii*) If $(f \circ \vartheta - f)^+ \in \mathrm{L}^1$, then the representation

$$f \circ \vartheta^n = f + \sum_{i=0}^{n-1} (f \circ \vartheta - f) \circ \vartheta^i$$

and the (extended version of the) ergodic theorem say that $\lim_{n+\infty} \left(\frac{1}{n} f \circ \vartheta^n\right) =: z$ exists (and is possibly $-\infty$) with $\mathbb{E}(z) = \mathbb{E}(f \circ \vartheta - f)$ (possibly $-\infty$). By (*ii*) this limit is necessarily equal to 0 $\mathbb{P}$-a.s. Hence $\mathbb{E}(z) = \mathbb{E}(f \circ \vartheta - f) = 0$, in particular $f \circ \vartheta - f \in \mathrm{L}^1$. Analogously for the case $(f \circ \vartheta - f)^- \in \mathrm{L}^1$.

□

*Proof of the MET for one-sided discrete time.*

(A) Under the asssumption $\log^+ \|A\| \in \mathrm{L}^1$, the first condition of the deterministic MET is true for the sequence $A_n = \mathcal{A}^1_{\Theta^{n-1}(\omega,x)}$ for $(\omega, x)$ in an invariant set $\Omega_1$ of full measure, by Lemma 19 (*i*) applied to $f(\omega, x) = \log \left\|\mathcal{A}^1_{\omega,x}\right\|$.

Under the same assumption, the Lemma 14 (*A*), hence in particular the second condition of the deterministic MET holds on a forward invariant set $\Omega_2$ of full measure.

Consequently, (*i*), (*ii*), (*iii*) and (*v*) are true on $\mathrm{U} := \Omega_1 \cap \Omega_2$ which is forward invariant and full.

For (*iv*) just note that $\mathcal{A}^n_{\Theta(\omega,x)} \mathcal{A}^1_{\omega,x} = \mathcal{A}^{n+1}_{\omega,x}$ which immediately implies $\lambda(\Theta(\omega, x), \mathcal{A}^1_{\omega,x} v) = \lambda(\omega, x, v)$. Take $x \in \mathrm{F}_i(\omega, x)$ which by (*iii*) is equivalent to $\lambda(\omega, x, v) \leqslant \lambda_i(\omega, x)$. Hence

$$\lambda(\Theta(\omega, x), \mathcal{A}^1_{\omega,x} v) = \lambda(\omega, x, v) \leqslant \lambda_i(\omega, x) = \lambda_i(\Theta(\omega, x)),$$

entailing $\mathcal{A}^1_{\omega,x} v \in \mathrm{F}_i(\Theta(\omega, x))$, or $\mathcal{A}^1_{\omega,x} \mathrm{F}_i(\omega, x) \subset \mathrm{F}_i(\Theta(\omega, x))$.

(B) This case follows from the fact that the the Lemma 14 (*B*) furnishes an invariant set of full measure on which all statements are true and all limits are finite (and constant in the ergodic case).

As for the invariance of $\mathrm{F}_i(\omega, x)$, note that since $A(\omega, x)$ is non-singular, $\dim A(\omega, x)\mathrm{F}_i(\omega, x) = \dim \mathrm{F}_i(\omega, x)$. Hence (*A*)(*iv*) and $\dim \mathrm{F}_i(\omega, x) = \dim \mathrm{F}_i(\Theta(\omega, x))$ imply $A(\omega, x)\mathrm{F}_i(\omega, x) = \mathrm{F}_i(\Theta(\omega, x))$.

□

*Proof of the MET.* The continuous time case can be reduced to the discrete time case as follows:

(1) The integrability conditions assure that the corresponding continuous time Furstenberg-Kesten theorem is valid on a set U of full measure which is $\Theta^t$-invariant.

(2) For fixed $(\omega, x) \in \mathrm{U}$ we introduce, as for $\mathrm{T} = \mathbf{N}$, the flag $\mathrm{F}(t, \omega, x)$ corresponding to $\left((\mathcal{A}^t_{\omega,x})^\top \mathcal{A}^t_{\omega,x}\right)^{1/2}$ and show that

– $(\mathrm{F}(t,\omega,x))_{t\geqslant 0}$ converges in $\mathrm{F}_\tau(d)$ to $\mathrm{F}(\omega,x) := \lim_{n+\infty} \mathrm{F}(n,\omega,x)$ (the latter limit is known to exist on U),

– the convergence is exponentially fast,

$$\limsup_{t+\infty}\left(\frac{1}{t}\log\delta(\mathrm{F}(t,\omega,x),\mathrm{F}(\omega,x))\right) \leqslant -h, \tag{5}$$

from which all other steps of the proof follow.

(3) The claims in (2) would follow from

$$\limsup_{n+\infty}\left(\frac{1}{n}\log\delta(\mathrm{F}(n,\omega,x),\mathrm{F}(n+s,\omega,x))\right) \leqslant -h \text{ uniformly in } s\in[0,1]. \tag{6}$$

Writing $n+s=t$, $n=[t]$ in (6) we obtain

$$\limsup_{t+\infty}\left(\frac{1}{t}\log\delta\left(\mathrm{F}([t],\omega,x),\mathrm{F}(t,\omega,x)\right)\right) \leqslant -h,$$

in particular $\mathrm{F}(t,\omega,x)\longrightarrow\mathrm{F}(\omega,x)$, and (5) follows from the triangle inequality.

(4) All that remains to be proved is (6). An inspection of the proof of Lemma 17 (step 3, case 1 and case 2(a)) shows that we have to assure that the right-hand side of

$$\|P_j(n+t,\omega,x)P_i(n,\omega,x)\| \leqslant \begin{cases} \left\|\mathcal{A}^t_{\Theta^n(\omega,x)}\right\|\dfrac{\sigma_i(\mathcal{A}^n_{\omega,x})}{\sigma_j(\mathcal{A}^{n+t}_{\omega,x})}, & \text{if } i>j,\\[2ex] \left\|(\mathcal{A}^t_{\Theta^n(\omega,x)})^{-1}\right\|\dfrac{\sigma_j(\mathcal{A}^{n+t}_{\omega,x})}{\sigma_i(\mathcal{A}^n_{\omega,x})}, & \text{if } i<j, \end{cases}$$

decays to 0 with exponential speed $-|\lambda_i-\lambda_j|$ uniformly with respect to $t\in[0,1]$. This is the case provided

$$\limsup_{n+\infty}\left(\frac{1}{n}\sup_{0\leqslant t\leqslant 1}\log\left\|(\mathcal{A}^t_{\Theta^n(\omega,x)})^{\pm 1}\right\|\right) = \limsup_{n+\infty}\left(\frac{1}{n}\alpha^{\pm}(\theta(n)\omega)\right) \leqslant 0$$

(note that since $\mathcal{A}^0_{\omega,x}=\mathrm{id}$ : $\sup\log=\sup\log^+$). The integrability conditions $\alpha^{\pm}\in\mathrm{L}^1$ and the ergodic theorem assure that in fact $\lim_{n+\infty}\alpha^{\pm}(\theta(n)\omega)/n=0$ on a set of full measure which is invariant with respect to $\Theta^1$. In step (*iii*) of the proof of Furstenberg-Kesten Theorem we showed that this set is even $\Theta^t$-invariant, and its defining property was built into the $\Theta^t$-invariant set U on which we are working.

□

**Proposition 20** (Lyapunov spectrum)**.** *We have that the quantities from Lemma 17 (A) coincides with the Lyapunov spectrum of the cocycle $\mathcal{A}$:*

$$\mathscr{S}(\theta,\mathcal{A}) = \{(\lambda_i(\cdot),d_i(\cdot)),\ i\in\{1,\dots,p(\cdot)\}\}$$

*Proof.* This proposition is clear as we have constructed the objects in Lemma 17 (*A*) by the Furstenberg-Kesten theorem on which Definition of Lyapunov spectrum is based. □

**Properties**

**Corollary 21.** *We have that:*

$$\forall\, v \in \mathbf{R}^d \backslash \{0\} \;:\quad \lambda(\Theta^t(\omega,x), \mathcal{A}^t_{\omega,x} v) \;=\; \lambda(\omega,x,v),$$

*whence*

$$\forall\, i \in \{1,\dots,p(\omega,x)\} \;:\quad \mathcal{A}^t_{\omega,x} \mathrm{F}_i(\omega,x) \;\subset\; \mathrm{F}_i(\Theta^t(\omega,x)).$$

*In the ergodic case:*

$$\forall\, i \in [\![1,d]\!], \quad \forall\, (\omega,x) \in \mathrm{U}, \quad \forall\, t \,>\, 0 \;:\quad \mathcal{A}^t_{\omega,x} \mathrm{F}_i(\omega,x) \;=\; \mathrm{F}_i(\Theta^t(\omega,x)).$$

*Remark.* The previous lemma provides that:

$$\lambda_1 \;:=\; \lim_{t+\infty} \frac{1}{t} \log \left\| \mathcal{A}^t_{\omega,x} \right\|, \quad \lambda_r \;:=\; \lim_{t+\infty} -\frac{1}{t} \log \left| (\mathcal{A}^t_{\omega,x})^{-1} \right|.$$

Indeed, for any $A \in \mathrm{GL}_d(\mathbf{R})$ we have:

$$\sigma_1(A) \;=\; |A| \qquad \text{and} \qquad \sigma_d(A) \;=\; |A^{-1}|^{-1}.$$

We then denote $\lambda^+ = \lambda_1$ and $\lambda^- = \lambda_r$. In particular, we have:

$$\lambda^+ \,>\, \lambda^- \qquad \Longleftrightarrow \qquad r \,>\, 1,$$

*i.e.*, there exist at least two distinct Lyapunov exponents if and only if $\lambda^+ > \lambda^-$.

**Proposition 22** (Measurability)**.** *We denote $G_k(d)$ the Grassmann manifold of $k$-dimensional subspaces of $\mathbf{R}^d$ (measurably extended to $\{\omega : p(\omega,x) \geqslant i\} \in \mathcal{F}$).*

- $p : \Omega \times \mathrm{X} \longrightarrow [\![1,d]\!]$ *(measurably extended from* U *to* $\Omega \times \mathrm{X}$*),*
- $\lambda_i : \{(\omega,x) : p(\omega,x) \,\geqslant\, i\} \longrightarrow \mathbf{R} \cup \{-\infty\}$*,*
- $d_i : \{(\omega,x) : p(\omega,x) \,\geqslant\, i\} \longrightarrow [\![1,d]\!]$*,*
- $\mathrm{E}_i : \{(\omega,x) : p(\omega,x) \,\geqslant\, i\} \longrightarrow \bigcup_{k=1}^d G_k(d)$*,*
- $\mathrm{F}_i : \{(\omega,x) : p(\omega,x) \,\geqslant\, i\} \longrightarrow \bigcup_{k=1}^d G_k(d)$*,*

*are measurable.*

*Proof.* The measurability of $p$, $\lambda_i$ and $d_i$ is a consequence of the measurability of the $\gamma^{(i)}$ which is assured by the Furstenberg-Kesten theorem. There remains to be considered the measurability of $\mathrm{E}_i$ and $\mathrm{F}_i$. The following will become clear in the proof of Proposition 18: on the measurable set $\mathrm{U}_{i,k} := \{(\omega,x) \in \mathrm{U} : d_i(\omega,x) = k\}$, $\mathrm{E}_i(\omega,x)$ will be the limit of a sequence $\mathrm{E}_i(n,\omega,x)$ of measurable functions with values in the Grassmannian $G_k(d)$ and hence measurable itself, similarly for $\mathrm{F}_i$. □

### 3.5. Particular cases

**Projective RDS associated to the cocycle $\mathcal{A}$**

Let us write $\mathrm{P}^{d-1}$ for the projective space associated to $\mathbf{R}^d$. The action of an invertible matrix $A \in \mathrm{GL}_d(\mathbf{R})$ descends to a well-defined action

$$\mathrm{A} : \mathrm{P}^{d-1} \longrightarrow \mathrm{P}^{d-1}.$$

Hence, we can think of the cocycle $\mathcal{A}$ as giving rise to an RDS on the product $\mathrm{X} \times \mathrm{P}^{d-1}$:

**Definition 27.** We call **projective** RDS associated to $\mathcal{A}$ every RDS $(\mathcal{P}_\omega)_{\omega\in\Omega}$ defined on $\mathrm{X}\times\mathrm{P}^{d-1}$ for every $\omega\in\Omega$ by

$$\begin{array}{rccl}\mathcal{P}_\omega: & \mathrm{T}\times(\mathrm{X}\times\mathrm{P}^{d-1}) & \longrightarrow & \mathrm{X}\times\mathrm{P}^{d-1}\\ & (t,x,v) & \longmapsto & \mathcal{P}^t_\omega(x,v)\,=\,(\mathcal{T}^t_\omega x,\,\mathcal{A}^t_{\omega,x}v),\end{array}$$

**Proposition 23.** *Every projective RDS is continuous on* $\mathrm{X}\times\mathrm{P}^{d-1}$.

*Proof.* Straightforward. □

The following condition is the independent increments condition for projective RDS:

**Definition 28** (Projective independent increments condition)**.** We say that a projective RDS $\mathcal{P}$ have **projective independent increments** if for all $s,t>0$:

$$(\mathcal{T}^t_{\cdot},\mathcal{A}^t_{\cdot})\quad\text{and}\quad(\mathcal{T}^s_{\theta^t\cdot},\mathcal{A}^s_{\theta^t\cdot,\cdot})\quad\text{on }(\Omega,\mathcal{F},\mathbb{P})\text{ are independent,}$$

where the variables are $\mathcal{C}_{u,b}(\mathrm{X},\mathrm{X})\times\mathcal{C}_{u,b}(\mathrm{X},\mathcal{M}_d(\mathbf{R}))$-valued random variables.

**Proposition 24.** *Let $\mathcal{P}$ be a projective RDS on $\mathrm{X}\times\mathrm{P}^{d-1}$ which has projective independent increments. Then, the stochastic process $(x_t,v_t)_{t\in\mathrm{T}}$ given by:*

$$\left\{\begin{array}{l}(x_t,v_t)\,=\,\mathcal{P}^t_\omega(x,v)\,=\,(\mathcal{T}^t_\omega x,\,\mathcal{A}^t_{\omega,x}v)\\ (x_0,v_0)\,=\,(x,v)\end{array}\right.$$

*is a Markov process with transition kernel defined for $((x,v),\mathrm{K})\in(\mathrm{X}\times\mathrm{P}^{d-1})\times\mathcal{B}_{(\mathrm{X}\times\mathrm{P}^{d-1})}$ by:*

$$\widehat{\mathrm{P}}_t((x,v),\mathrm{K})\;:=\;\mathbb{P}\left(\left\{\omega\in\Omega\,:\,(\mathcal{T}^t_\omega x,\mathcal{A}^t_{\omega,x}v)\in\mathrm{K}\right\}\right)$$

*Proof.* Straightforward by Lemma 5. □

**Proposition 25.** *The associated skew product semiflow is:*

$$\begin{array}{rccl}\tau: & \Omega\times\mathrm{T}\times\mathrm{X}\times\mathrm{P}^{d-1} & \longrightarrow & \Omega\times\mathrm{X}\times\mathrm{P}^{d-1}\\ & (\omega,t,x,v) & \longmapsto & \tau^t(\omega,x,v)\,=\,(\theta^t\omega,\mathcal{T}^t_\omega x,\mathcal{A}^t_{\omega,x}v).\end{array}$$

*Proof.* Straightforward. □

We now turn our attention to the relationship between the ergodic theory of the projective process and the MET. It is not hard to see that any stationary measure $\nu$ for $(\widehat{\mathrm{P}}_t)$ must project to some $(\mathrm{P}_t)$-stationary measure $\mu$ on the X-factor.

**Lemma 26.** *Given a stationary measure $\mu$ for $(\mathrm{P}_t)$, there exists at least one stationary measure $\nu$ for the projective semigroup $(\widehat{\mathrm{P}}_t)$ such that*

$$\nu(\mathrm{A}\times\mathrm{X})\,=\,\mu(\mathrm{A}).$$

*Proof.* By Lemma 9. □

If $\nu$ as above is the unique stationary measure with marginal $\mu$, then we obtain the following refinement of the MET.

**Proposition 27** (MET for projective RDS)**.** *Assume that there is only one stationary measure $\nu$ for the projective RDS projecting to $\mu$ on the* X*-factor. Then, we have that:*

$$(\mathbb{P}\times\mu)\text{-}\forall\exists\,(\omega,x)\in\Omega\times\mathrm{X},\quad\forall\,v\in\mathbf{R}^d\backslash\{0\}\;:\quad\lim_{t+\infty}\frac{1}{t}\log\left\|\mathcal{A}^t_{\omega,x}v\right\|\,=\,\lambda_1.$$

*Proof.* Straightforward. □

**Matrix RDS associated to the cocycle $\mathcal{A}$**

The cocycle $\mathcal{A}$ also gives rise to an RDS on the product space $\mathrm{X} \times \mathcal{M}_d(\mathbf{R})$:

**Definition 29.** We call **matrix** random dynamical system associated to $\mathcal{A}$ every RDS $(\mathcal{M}_\omega)_{\omega\in\Omega}$ defined on $\mathrm{X} \times \mathcal{M}_d(\mathbf{R})$ for all $\omega \in \Omega$ by

$$\begin{array}{rccc} \mathcal{M}_\omega : & \mathrm{T} \times (\mathrm{X} \times \mathcal{M}_d(\mathbf{R})) & \longrightarrow & \mathrm{X} \times \mathcal{M}_d(\mathbf{R}) \\ & (t, x, B) & \longmapsto & \mathcal{M}^t_\omega(x, B) \,=\, (\mathcal{T}^t_\omega x,\, \mathcal{A}^t_{\omega,x} B), \end{array}$$

We can similarly define an independent increments condition for matrix RDS and then:

**Proposition 28.** *Let $\mathcal{M}$ be a matrix RDS on $\mathrm{X} \times \mathcal{M}_d(\mathbf{R})$ which has matrix independent increments. Then, the stochastic process $(x_t, B_t)_{t\in\mathrm{T}}$ given by:*

$$\begin{cases} (x_t, B_t) \,=\, \mathcal{P}^t_\omega(x, v) \,=\, (\mathcal{T}^t_\omega x,\, \mathcal{A}^t_{\omega,x} B) \\ (x_0, B_0) \,=\, (x, B) \end{cases}$$

*is a Markov process with transition kernel defined for $((x, B),\, \mathrm{K}) \in (\mathrm{X} \times \mathcal{M}_d(\mathbf{R})) \times \mathcal{B}_{(\mathrm{X}\times\mathcal{M}_d(\mathbf{R}))}$ by:*

$$\mathrm{Q}_t((x, B),\, \mathrm{K}) \;:=\; \mathbb{P}\left(\left\{\omega \in \Omega \,:\, (\mathcal{T}^t_\omega x, \mathcal{A}^t_{\omega,x} B) \in \mathrm{K}\right\}\right)$$

*Proof.* Straightforward. □

*Remark.* Note that if $B \in \mathrm{GL}_d(\mathbf{R})$ and $\mathrm{K} \,=\, \mathrm{K}_1 \times \mathrm{K}_2$ where $\mathrm{K}_1 \subset \mathrm{X}$, $\mathrm{K}_2 \subset \mathcal{M}_d(\mathbf{R})$, then

$$\mathrm{Q}_t((x, B), \mathrm{K}) \,=\, \mathrm{Q}_t\left((x, \mathrm{id}),\, \mathrm{K}_1 \times (\mathrm{K}_2 B^{-1})\right).$$

Thus, frequently we are only interested in the Markov kernel $(\mathrm{Q}_t)$ evaluated at $(x, \mathrm{id})$.

**Inverse-transpose cocycle associated to the cocycle $\mathcal{A}$**

Assume that $\mathcal{A}$ is invertible. Another important tool associated to the cocycle $\mathcal{A}$ is:

**Definition 30.** We call the **inverse-transpose** cocycle associated to $\mathcal{A}$, denoted by $(-\top)$-cocycle $\mathcal{A}$, the cocycle defined for $(\omega, t, x) \in \Omega \times \mathrm{T} \times \mathrm{X}$ by:

$$\breve{\mathcal{A}}^t_{\omega,x} \;:=\; \left(\mathcal{A}^t_{\omega,x}\right)^{-\top},$$

where "$-\top$" refers to the inverse-transpose of a matrix of $\mathrm{GL}_d(\mathbf{R})$.

**Proposition 29.** *$\breve{\mathcal{A}}$ is a linear cocycle over the RDS $\mathcal{T}$. Moreover, if $\mathcal{A}$ has independent increments or is integrable, then $\breve{\mathcal{A}}$ has independent increments or is integrable respectively.*

*Proof.* Straightforward. □

Therefore the MET and all the aforementioned material applies, and these objects can be directly represented in terms of the exponents and subspaces of the original cocycle $\mathcal{A}$:

**Proposition 30** (Inverse-transpose MET)**.** *Supposed that $\mathcal{T}$ has independent increments and $\mathcal{A}$ is invertible and integrable. Then $\breve{\mathcal{A}}$ has Lyapunov exponents*

$$\breve{\lambda}_1 \,>\, \cdots \,>\, \breve{\lambda}_{\breve{r}},$$

*and associated subspaces $\breve{\mathrm{F}}_2(\omega, x), \cdots, \breve{\mathrm{F}}_{\breve{r}}(\omega, x)$ where for almost every $(\omega, x) \in \Omega \times \mathrm{X}$:*

$$\breve{r} \,=\, r, \qquad \breve{\lambda}_i \,=\, -\lambda_{r-(i-1)} \qquad \text{and} \qquad \breve{\mathrm{F}}_i(\omega, x) \,=\, (\mathrm{F}_{r-(i-1)+1}(\omega, x))^{\perp}.$$

*Proof.* This follows on applying the lemma of the MET to the cocycle $\breve{\mathcal{A}}$ and noting that for all $(\omega, x) \in \Omega \times \mathrm{X}$ and $t \in \mathrm{T}$:

$$\log((\breve{\mathcal{A}}^t_{\omega,x})^\top \breve{\mathcal{A}}^t_{\omega,x}) \;=\; -\log((\mathcal{A}^t_{\omega,x})^\top \mathcal{A}^t_{\omega,x}).$$

□

*Remark.* If the cocycle $\mathcal{A}$ has projective independent increments, then $\breve{\mathcal{A}}$ induces the $(-\top)$-projective process $(x_t, \breve{v}_t)$ on $\mathrm{X} \times \mathrm{P}^{d-1}$ defined for fixed initial $x_0 \in \mathrm{X}$, $\breve{v}_0 \in \mathrm{P}^{d-1}$ by setting $\breve{v}_t$ to be the projective representative of $\breve{\mathcal{A}}_{\omega,x_0}\breve{v}_0$. Then, all the material from projective process applies with $\breve{\mathcal{A}}$ replacing $\mathcal{A}$ and $(x_t, \breve{v}_t)$ replacing $(x_t, v_t)$. In particular, the conclusions of the proposition related to the MET for projective RDS hold with $\breve{\mathcal{A}}$ replacing $\mathcal{A}$ when the stationary measure for $(x_t, \breve{v}_t)$ projecting to $\mu$ on the X factor is unique.

## 4. **Positiveness of Lyapunov exponent**

This section is fundamental to study the Lagrangian chaos. It is really hard to prove that a system has a positive Lyapunov exponent. For instance, in deterministic systems, it is almost impossible to tell if there's a exponent even for very simple systems. In the random case, we will use a key idea which comes from RDS with à la Furstenberg rigidity: it says that if you have a volume preserving system, then in order to not being positive you need an almost-surely invariant structure. Hence, we will establish a new refinement of Furstenberg's criterion. Let:

- $(\mathrm{X}, d)$ be a separable, complete and metric space,
- $(\Omega, \mathcal{F}, \mathbb{P}, (\theta^t)_{t\in\mathrm{T}})$ be a measure-preserving system,
- $\mathcal{T}$ be a strongly continuous RDS with independent increments, and which admits:
- $\mu$ be a stationary measure for $\mathcal{T}$.

### 4.1. **Fiber measures**

**Definition 31.** We say that a measure $\nu$ is a **fiber measure** if:

$$\nu \text{ is a probability measure on } \mathrm{P}^{d-1}.$$

*Remark.* For a family of fiber measures $(\nu_x)$, we call the *fiber measure at* $x$ as the individual measure $\nu_x$.

**Definition 32.** Let $A \in \mathcal{M}_d(\mathbf{R})$. We denote the corresponding map on projective space as:

$$\mathrm{A} : \mathrm{P}^{d-1} \longrightarrow \mathrm{P}^{d-1}.$$

Let $\nu$ be a fiber measure. Then, the **pushforward** of $\nu$ by $A$ is:

$$A_*\nu \;:=\; \nu \circ \mathrm{A}^{-1}.$$

**Definition 33.** We say that the family of fiber measures $(\nu_x)$ is **invariant** if:

$$\forall\, t \in \mathrm{T}, \quad (\mathbb{P}\times\mu)\text{-}\forall\exists\,(\omega, x) \in \Omega \times \mathrm{X} \;:\quad (\mathcal{A}^t_{\omega,x})_*\nu_x \;=\; \nu_{(\mathcal{T}^t_\omega x)}.$$

*Remark.* Hence, a family of fiber measures $(\nu_x)$ is invariant iff for every test function $\varphi \in \mathscr{D}(\mathrm{P}^{d-1})$:

$$\int_{\mathrm{P}^{d-1}} \varphi(s)\,\mathrm{d}\nu_{\mathcal{T}^t_{\omega,x}}(s) \;=\; \int_{\mathrm{P}^{d-1}} \varphi(\mathcal{A}^t_{\omega,x}s)\nu_x(\mathrm{d}s).$$

4.2. **Furstenberg's criterion**

Assume that the stationary measure $\mu$ is ergodic and recall that $(\Theta^t)$ is a measurable measure-preserving semiflow. Let $\mathcal{A}$ be invertible, integrable and have independent increments. From the MET, the following limits exists:

$$\lambda_+ \,=\, \lim_{t+\infty}\left(\frac{1}{t}\mathbb{E}\left(\log\left\|\mathcal{A}^t_{(\cdot,\cdot)}\right\|\right)\right) \qquad \text{and} \qquad \lambda_- \,=\, \lim_{t+\infty}\left(-\frac{1}{t}\mathbb{E}\left(\log\left\|(\mathcal{A}^t_{(\cdot,\cdot)})^{-1}\right\|\right)\right).$$

*Remark.* Since for all $A \in \mathrm{GL}_d(\mathbf{R}) \,:\, \|A\|\|A^{-1}\| \geqslant 1$, we have: $\lambda_+ \,\geqslant\, \lambda_-$. We want to study under which conditions equality may hold.

NOTATIONS 6. Let $\mu$ and $v$ be nonnegative finite (not necessarily probability) measures on some measurable space. We denote $\mathrm{d}\mu/\mathrm{d}\nu$ the density of the absolutely continuous part in the Lebesgue decomposition of $\mu$ with respect to $\nu$, that is it's the $\nu$-a.s. unique function $f$ such that

$$f(s) \,=\, \frac{\mathrm{d}\mu}{\mathrm{d}\nu}(s) \qquad \Longleftrightarrow \qquad \mu \,=\, f\nu + \gamma,$$

with $\gamma$ and $\nu$ mutually singular. In particular, we have $\nu$-a.s.: $\mathrm{d}\mu/\mathrm{d}v \,\geqslant\, 0$.

**Definition 34.** We define the **relative entropy** of $\mu$ with respect to $\nu$ by:

$$h(\mu \mid \nu) \,:=\, -\int \log\left(\frac{\mathrm{d}\mu}{\mathrm{d}\nu}(s)\right)\mathrm{d}\nu(s).$$

**Definition 35.** We define the **entropy** of a $\tau$-invariant measure $\nu$ by:

$$\alpha_\nu(t) \,:=\, \mathbb{E}_{(\mathbb{P}\times\mu)}\left[h\left((\mathcal{A}^t_{\omega,x})^{-1}\nu_{\mathcal{T}^t_\omega x} \mid \nu_x\right)\right].$$

**Proposition 31.** *We have $\alpha_\nu \,\geqslant\, 0$. Equality $\alpha_\nu \,=\, 0$ holds iff for $(\mathbb{P}\times\mu)$-a.e. $(\omega, x)$:*

$$(\mathcal{A}^t_{\omega,x})_*\nu_x \,=\, \nu_{(\mathcal{T}^t_\omega x)}$$

*Proof.* As $(-\log)$ is strictly convex, then from Jensen's inequality:

$$\int -\log\left(\frac{\mathrm{d}}{\mathrm{d}\nu_x}(\mathcal{A}^t_{\omega,x})^{-1}\nu_{\mathcal{T}^t_\omega x}(s)\right)\mathrm{d}\nu_x(s) \,\geqslant\, -\log\left(\int \frac{\mathrm{d}}{\mathrm{d}\nu_x}(\mathcal{A}^t_{\omega,x})^{-1}\nu_{\mathcal{T}^t_\omega x}(s)\mathrm{d}\nu_x(s)\right) \,\geqslant\, 0.$$

Moreover, we have:

$$\frac{\mathrm{d}}{\mathrm{d}\nu_x}(\mathcal{A}^t_{\omega,x})^{-1}\nu_{\mathcal{T}^t_\omega x} \text{ is } \nu_x\text{-constant} \quad\Longleftrightarrow\quad (\mathcal{A}^t_{\omega,x})^{-1}\nu_{\mathcal{T}^t_\omega x} \,=\, \nu_x \quad\Longleftrightarrow\quad \nu_{(\mathcal{T}^t_\omega x)} \,=\, \mathcal{A}^t_{\omega,x}\cdot\nu_x.$$

□

**Lemma 32.** *We have for all $\omega \in \Omega$:*

$$\alpha_\nu \,\leqslant\, (d-1)\mathbb{E}_\mu\left(\log\left\|\mathcal{A}^1_{(\omega,\cdot)}\right\| + \log\left\|(\mathcal{A}^1_{(\omega,\cdot)})^{-1}\right\|\right).$$

*Proof.* This lemma is proved in [23]. □

**Theorem 33.** *Let $\nu$ a $\tau$-invariant measure on $\mathrm{X}\times\mathrm{P}^{d-1}$, then :*

$$\alpha_\nu \,\leqslant\, (d-1)(\lambda^+ - \lambda^-).$$

*Proof.* Let's apply the previous proposition to every $\tau^n$ (with $n > 0$) and $\nu$, hence:

$$\alpha_\nu(n) \,\leqslant\, (d-1)\mathbb{E}\left(\log\left\|\mathcal{A}^n_{(\cdot,\cdot)}\right\| + \log\left\|(\mathcal{A}^n_{(\cdot,\cdot)})^{-1}\right\|\right).$$

However, using that for $n > 0 : \alpha_\nu(n) \,=\, n\alpha_\nu(1)$, and the definition of $\lambda^+$ and $\lambda^-$, the theorem follows. □

**Theorem 34** (Furstenberg's criterion [23])**.** *If* $\lambda^+ = \lambda^-$*, then:*

- *there exists a family of Borel measures* $(\nu_x)$ *on* $\mathrm{P}^{d-1}$ *such that* $x \longmapsto \nu_x$ *is measurable,*
- $\forall\, t \in \mathrm{T}, \quad (\mathbb{P}\times\mu)\text{-}\forall\exists\,(\omega,x) \in \Omega\times\mathrm{X} \;:\quad \mathcal{T}^t_\omega x \in \mathrm{X} \qquad \text{and} \qquad (\mathcal{A}^t_{\omega,x})_*\nu_x = \nu_{(\mathcal{T}^t_\omega x)}.$

*Proof.* The equality of Lyapunov exponents implies by Theorem 33 that the entropy is zero. Thus the theorem follows from Proposition 31. We pass from the discrete-time setting above to our present continuous-times setting and that gives us the $(\mathbb{P}\times\mu)$-almost sure set may depend on $t$. □

*Remark.* It says that if $\lambda^+ = \lambda^-$, then there is a deterministic (*i.e.*, $\omega$-independent) structure preserved by the cocycle $\mathcal{A}$ with probability one.

**Corollary 35.** *If* $\lambda^+ = \lambda^-$*, the measure* $\nu$ *defined on* $\mathrm{X}\times\mathrm{P}^{d-1}$ *by:*

$$\forall\,(x,v) \in \mathrm{X}\times\mathrm{P}^{d-1} \;:\quad \mathrm{d}\nu(x,v) = \mathrm{d}\mu(x)\,\mathrm{d}\nu_x(v),$$

*is a stationary measure for* $(\widehat{\mathrm{P}}^t)$ *associated to the projective RDS on* $\mathrm{X}\times\mathrm{P}^{d-1}$*.*

### 4.3. **A la Furstenberg theorem: classification of invariant fiber measure families**

In this section we establish a new refinement of Furstenberg's criterion which is a classification of the fiber measures $(\nu_x)_{x\in\mathrm{supp}\,\mu}$ comprising a family satisfying the previous *topological* relation. Leveraging the continuity result, we take advantage of algebraic properties of $\mathrm{SL}_d(\mathbf{R})$ to obtain a classification for the family $(\nu_x)$ under the assumption that $\lambda^+ = \lambda^-$ as in Furstenberg's criterion.

#### Continuous dependence of an invariant fiber measure family

First, we establish the continuous dependence of an invariant fiber measure family $(\nu_x)$ on the base point $x$ under the assumption that the Markov semigroup $\mathrm{P}_t$ associated to the RDS $\mathcal{T}$ has the strong Feller property.

**Definition 36.** We say that the Markov semigroup $(\mathrm{P}_t)$ is:

- **strong Feller** if for all $t \in \mathrm{T} \;:\quad \forall\, h \in \mathrm{L}^\infty(\mathrm{X},\mathbf{R}), \quad \mathrm{P}_t h \in \mathcal{C}_b(\mathrm{X},\mathbf{R})$.
- **ultra Feller** if for all $t \in \mathrm{T} \;:\quad x \longmapsto \mathrm{P}_t(x,\cdot)$ is continuous, in the total variation distance $\mathrm{d}_{TV}$ on the space of finite signed measures on X.

**Lemma 36.** *Assume* X *is a Polish space. Then:*

- *Strong Feller on* X $\quad\Longrightarrow\quad$ *Ultra Feller on* X*,*
- *for all stationary measure* $\mu$ *for* $(\mathrm{P}_t)$ *and* $\mathrm{K}\subset\mathrm{X}$ *a Borel set of full* $\mu$ *measure, then:*

$$\forall\, t \in \mathrm{T}, \quad \forall\, x \in \mathrm{supp}\,(\mu) \;:\quad \mathrm{P}_t(x,\mathrm{K}) = 1.$$

*Proof.*

- It is sufficient to use the dual representation of the total variation:

$$\|\mu\|_{TV} = \sup\left(\int_{\mathrm{X}} f\,\mathrm{d}\mu,\; f \in \mathcal{C}_c(\mathrm{X}) \;:\; |f| \leqslant 1\right).$$

- First, we check that for all $t \in \mathrm{T}$: $\{x \in \mathrm{X} \;:\; \mathrm{P}_t(x,\mathrm{K}) = 1\}$ is dense in $\mathrm{supp}\,(\mu)$. Hence, the result follows from continuity in total variation.

□

We want to turn the measurable information contained in the Furstenberg's criterion into topological information. This will be accomplished in two phases: the first step is a continuity result for the invariant measure family $(\nu_x)_{x\in \mathrm{X}}$:

**Proposition 37** (First step)**.** *Assume* $(\mathrm{P}_t)$ *is strong Feller, and let* $(\nu_x)_{x\in\mathrm{X}}$ *be an invariant fiber measure family. Then, there exists an invariant fiber measure family* $(\bar{\nu}_x)_{x\in\mathrm{supp}\,(\mu)}$ *such that:*

- $(\bar{\nu}_x)$ *is a* $\mu$*-a.s version of the original family* $(\nu_x)$*:*

$$\mu\text{-}\forall\exists\, x\in\mathrm{supp}\,(\mu)\ :\quad \nu_x\,=\,\bar{\nu}_x.$$

- $(\bar{\nu}_x)$ *is continuously varying in the weak-*$*$ *topology on* $\mathrm{P}^{d-1}$*.*

*Remark.* This proposition let us replace the possibly discontinuous invariant measure family $(\nu_x)$ with a continuously varying invariant measure family $(\bar{\nu}_x)$ defined at each $x\in\mathrm{supp}\,(\mu)$ (at the expense of modifying $(\nu_x)$ on a set of $\mu$-measure zero).

**Lemma 38.** *For every* $g\in\mathcal{C}^\circ(\mathrm{P}^{d-1},\mathbf{R})$ *we define the following operator:*

$$\begin{array}{rccc} G: & \mathrm{X} & \longrightarrow & \mathbf{R} \\ & x & \longmapsto & \langle\nu_x,g\rangle\,=\displaystyle\int_{\mathrm{P}^{d-1}} g(v)\,\mathrm{d}\nu_x(v). \end{array}$$

*Hence, for all* $g\in\mathcal{C}^\circ(\mathrm{P}^{d-1},\mathbf{R})$*, there is a full* $\mu$*-measure subset* $\widetilde{\mathrm{X}}\subset\mathrm{supp}\,(\mu)$ *such that:*

$$\forall\,(x_n)_{n\geqslant 1}\in(\widetilde{\mathrm{X}})^{\mathbf{N}}\ \text{ Cauchy sequence}\quad:\quad (G(x_n))_{n\geqslant 1}\ \text{ is Cauchy sequence.}$$

*Proof.* Since $(\nu_x)$ is an invariant fiber measure family, it is straightforward to construct a full $\mu$-measure subset $\widetilde{\mathrm{X}}\subset\mathrm{supp}\,(\mu)$ such that:

$$\forall\, t\in\mathbf{Q}_+,\quad \forall\, x\in\widetilde{\mathrm{X}}\ :\quad \mathcal{T}^t_\omega x\in\widetilde{\mathrm{X}}\qquad\text{and}\qquad (\mathcal{A}^t_{\omega,x})_*\nu_x\,=\,\nu_{(\mathcal{T}^t_\omega x)}\ \text{ with probability 1.}$$

We denotes $\widehat{\mathrm{P}}_t$ and $\widetilde{\mathrm{P}}_t$ as the Markov semigroups associated respectively to the projective process and $\mathcal{A}$. Note that:

$$\forall\,\mathrm{K}\subset\mathrm{P}^{d-1},\quad \forall\,(x,v)\in\mathrm{X}\times\mathrm{K}\ :\quad \widehat{\mathrm{P}}_t((x,v),\mathrm{X}\times\mathrm{K})\,=\,\widetilde{\mathrm{P}}_t(v,\mathrm{K}).$$

1. Fix $x\in\widetilde{\mathrm{X}}$, let's integrate $(\mathcal{A}^t_{\omega,x})_*\nu_x\,=\,\nu_{(\mathcal{T}^t_\omega x)}$ with respect to $\mathrm{d}\mathbb{P}(\omega)$: hence, for all $g\in\mathcal{C}^\circ(\mathrm{P}^{d-1},\mathbf{R})$ we have

$$\int_\Omega\int_{\mathrm{P}^{d-1}} g(s)(\mathcal{A}^t_{\omega,x})_*\nu_x(\mathrm{d}s)\,\mathrm{d}\mathbb{P}(\omega)\,=\,\int_\Omega\int_{\mathrm{P}^{d-1}} g(s)\,\nu_{(\mathcal{T}^t_\omega x)}(\mathrm{d}s)\,\mathrm{d}\mathbb{P}(\omega).$$

For the left-hand side we have:

$$\begin{aligned}
\int_\Omega\int_{\mathrm{P}^{d-1}} g(s)(\mathcal{A}^t_{\omega,x})_*\nu_x(\mathrm{d}s)\,\mathrm{d}\mathbb{P}(\omega) &= \int_\Omega\int_{\mathrm{P}^{d-1}} g(s)\,\mathrm{d}\nu_x\left((\mathcal{A}^t_{\omega,x})^{-1}s\right)\,\mathrm{d}\mathbb{P}(\omega)\\
&\underset{v=(\mathcal{A}^t_{\omega,x})^{-1}s}{=} \int_\Omega\int_{\mathrm{P}^{d-1}} g(\mathcal{A}^t_{\omega,x}v)\,\mathrm{d}\nu_x(v)\,\mathrm{d}\mathbb{P}(\omega)\\
&\underset{\nu_x=\bar{\nu}_x}{=} \int_{\mathrm{P}^{d-1}}\int_\Omega g(\mathcal{A}^t_{\omega,x}v)\mathrm{d}\mathbb{P}(\omega)\,\mathrm{d}\bar{\nu}_x(v)\\
&= \int_{\mathrm{P}^{d-1}} \widetilde{\mathrm{P}}_t g(v)\,\mathrm{d}\bar{\nu}_x(v)\\
&= \int_{\mathrm{P}^{d-1}}\left(\int_{\mathrm{P}^{d-1}} g(s)\,\widetilde{\mathrm{P}}_t(v,\mathrm{d}s)\right)\mathrm{d}\bar{\nu}_x(v)\\
&= \int_{\mathrm{P}^{d-1}}\left(\int_{\mathrm{X}\times\mathrm{P}^{d-1}} g(s)\,\widehat{\mathrm{P}}_t((x,v),\mathrm{d}y\mathrm{d}s)\right)\mathrm{d}\bar{\nu}_x(v)\\
&= \int_{\mathrm{P}^{d-1}} (\widehat{\mathrm{P}}_t g)(x,v)\,\mathrm{d}\bar{\nu}_x(v).
\end{aligned}$$

For the right-hand side, let's consider $h : y \in \mathrm{X} \longmapsto \displaystyle\int_{\mathrm{P}^{d-1}} g(s)\,\nu_y(\mathrm{d}s) \in \mathbf{R}$. Thus:

$$\begin{aligned}\int_\Omega \int_{\mathrm{P}^{d-1}} g(s)\,\nu_{(\mathcal{T}^t_\omega x)}(\mathrm{d}s)\,\mathrm{d}\mathbb{P}(\omega) &= \int_\Omega h(\mathcal{T}^t_\omega x)\,\mathrm{d}\mathbb{P}(\omega)\\ &= \mathrm{P}_t h(x)\\ &= \int_{\mathrm{X}} h(y)\mathrm{P}_t(x,\mathrm{d}y)\\ &= \int_{\mathrm{X}} \left(\int_{\mathrm{P}^{d-1}} g(s)\,\nu_y(\mathrm{d}s)\right)\mathrm{P}_t(x,\mathrm{d}y)\\ &= \int_{\mathrm{X}} G(y)\mathrm{P}_t(x,\mathrm{d}y)\\ &= \mathrm{P}_t G(x).\end{aligned}$$

Hence, for all $g \in \mathcal{C}^\circ(\mathrm{P}^{d-1},\mathbf{R})$: $\displaystyle\int_{\mathrm{P}^{d-1}} (\widehat{\mathrm{P}}_t g)(x,v)\,\mathrm{d}\bar\nu_x(v) \,=\, \mathrm{P}_t G(x)$.

2. Let $(x_n) \in (\widetilde{\mathrm{X}})^{\mathbf{N}}$ be a Cauchy sequence converging to some $x \in \mathrm{X}$. Fix $\varepsilon > 0$ and U a neighborhood of $x$. Without loss, $(x_n) \in \mathrm{U}^{\mathbf{N}}$. From Proposition 7:

$$\widehat{\mathrm{P}}_t g \;\xrightarrow[t\to 0^+]{}\; g \;\text{ uniformly on bounded subsets of } \mathrm{X}\times\mathrm{P}^{d-1},$$

we have that $\widehat{\mathrm{P}}_t g \;\xrightarrow[t\to 0^+]{}\; g$ uniformly on $\mathrm{U}\times\mathrm{P}^{d-1}$.

3. Fix $t_\varepsilon$ for which on all of $\mathrm{U}\times\mathrm{P}^{d-1}$ and for all $s \in [0,t_\varepsilon]$ : $|\widehat{\mathrm{P}}_s g - g| < \varepsilon$. Fix a rational $t_0 \in [0,t_\varepsilon]$. Given $n,m \geqslant 1$ we estimate:

$$\begin{aligned}|G(x_n) - G(x_m)| &= \left|\int g\,\mathrm{d}\nu_{x_n} - \int g\,\mathrm{d}\nu_{x_m}\right|\\ &\leqslant \int |g - \mathrm{P}_{t_0} g|\,\mathrm{d}\nu_{x_n} + \left|\int \mathrm{P}_{t_0} g\,\mathrm{d}\nu_{x_n} - \int \mathrm{P}_{t_0} g\,\mathrm{d}\nu_{x_m}\right| + \int |g - \mathrm{P}_{t_0} g|\,\mathrm{d}\nu_{x_m}\\ &\leqslant 2\varepsilon + |\mathrm{P}_{t_0} G(x_n) - \mathrm{P}_{t_0} G(x_m)|\,.\end{aligned}$$

Then, by the strong Feller property, $\mathrm{P}_{t_0} G$ is a continuous function. Hence, $(\mathrm{P}_{t_0} G(x_n))_n$ is a Cauchy sequence.

□

*Proof of the proposition.* Let us first define the family $(\bar\nu_x)$. Let $g \in \mathcal{C}^\circ(\mathrm{P}^{d-1},\mathbf{R})$ and let $G$ and $\widetilde{\mathrm{X}}$ be as the previous lemma.

1. For $x \in \widetilde{\mathrm{X}}$, we set: $\bar\nu_x \,:=\, \nu_x$. This ensures $(\bar\nu_x)$ is a version of $(\nu_x)$.

2. For each $x \in \mathrm{supp}\,(\mu)\backslash\widetilde{\mathrm{X}}$, and since $\widetilde{\mathrm{X}}$ is dense in $\mathrm{supp}\,(\mu)$, we can find a sequence $(x_n) \in (\widetilde{\mathrm{X}})^{\mathbf{N}}$ such that: $\lim_{n+\infty} x_n \,=\, x$.

3. From the previous lemma, $(G(x_n))$ is a Cauchy sequence, so by Prokhorov's Theorem[1], since $\mathrm{P}^{d-1}$ is compact, there exists a weak-$*$ limit:

$$\lim_{n+\infty} \bar\nu_{x_n} \,=:\, \bar\nu_x \,<\, +\infty,$$

and the definition of $\bar\nu_x$ is independent of $(x_n)$.

[1] Prokhorov's Theorem. If $(\nu_n)$ is tight and if each subsequence that converges weakly at all in fact converges weakly to $\nu$, then the entire sequence converges weakly to $\nu$.

This completes the definition of the family $(\bar{\nu}_x)$ and by construction, it is a $\mu$-almost-sure version of $(\nu_x)$.

To show continuity, we recall a characterisation for $(\mu_x)$ in the weak-$*$ topology on a topological space E:

$$(\mu_x) \text{ is continuously varying} \iff \forall\, \varphi \in \mathcal{C}^\circ(\mathrm{E}) \,:\quad x \longmapsto \int_{\mathrm{E}} \varphi \,\mathrm{d}\mu_x \text{ is continuous on E.}$$

Hence, we will check that

$$\bar{G} : x \longmapsto \int_{\mathrm{P}^{d-1}} g \,\mathrm{d}\bar{\nu}_x \text{ is continuous on } \mathrm{P}^{d-1}.$$

For this, fix $x \in \operatorname{supp}(\mu)$ and consider $(x_n) \in (\operatorname{supp}(\mu))^{\mathbf{N}}$ converging to $x$.

1. Since $\widetilde{\mathrm{X}}$ is dense in $\operatorname{supp}(\mu)$, for each $n$ we can fix $\tilde{x}_n \in \widetilde{\mathrm{X}}$ such that
$$\mathrm{d}(x_n, \tilde{x}_n) < \frac{1}{n} \qquad \text{and} \qquad \left|\bar{G}(x_n) - G(\tilde{x}_n)\right| < \frac{1}{n}.$$
2. Then,
$$\left|\bar{G}(x_n) - \bar{G}(x)\right| \leqslant \left|\bar{G}(x_n) - G(\tilde{x}_n)\right| + \left|G(\tilde{x}_n) - \bar{G}(x)\right| \leqslant \frac{1}{n} + \left|G(\tilde{x}_n) - \bar{G}(x)\right|.$$
The lemma and our definition of $\bar{\nu}_x$ imply that the second term goes to zero.

This completes the proof of continuity. □

**"Almost-sure" relation to "surely" relation**

Let us now describe the second step in our program, namely, turning the $\mathbb{P} \times \mu$-almost sure relation into an analogous relation holding for all $(\omega, x) \in \Omega \times \operatorname{supp}(\mu)$ and for all $t \in \mathrm{T}$, in a sense we make precise below.

Notations 7. We equipped the following with the product topology:

$$\mathcal{C} = \mathcal{C}_{u,b}(\mathrm{X}, \mathrm{X}) \times \mathcal{C}_{u,b}(\mathrm{X}, \mathcal{M}_d(\mathbf{R})).$$

where the elements of $\mathcal{C}$ are written $(T, A)$ where

$$T : \mathrm{X} \longrightarrow \mathrm{X} \qquad \text{and} \qquad A : \mathrm{X} \longrightarrow \mathcal{M}_d(\mathbf{R}).$$

Hence, we consider for all $t \in \mathrm{T}$ the $\mathcal{C}$-valued random variable:

$$\begin{array}{rccc} (\mathcal{T}^t, \mathcal{A}^t) : & (\Omega, \mathcal{A}) & \longrightarrow & \mathcal{C} \\ & \omega & \longmapsto & (\mathcal{T}^t_\omega, \mathcal{A}^t_{\omega,(\cdot)}). \end{array}$$

- The topological support of $(\mathcal{T}^t, \mathcal{A}^t)$ for $t \in \mathrm{T}$ and $\omega$ distributed as $\mathbb{P}$ is denoted:
$$\mathcal{S}_t = \bigcap_{\mathrm{F} \in \mathcal{F}} \mathrm{F},$$
where $\mathcal{F} = \left\{\mathrm{F} \subset \mathcal{C} \text{ closed} \,:\, \mathbb{P}\left((\mathcal{T}^t, \mathcal{A}^t) \in \mathrm{F}\right) = 1\right\}$.
- The closure of the union of the $\mathcal{S}_t$ in $\mathcal{C}$ is denoted: $\mathcal{S} = \overline{\bigcup_{t \in \mathrm{T}} \mathcal{S}_t}$.

**Proposition 39** (Second step)**.** $\forall\, x \in \mathrm{supp}\,(\mu),\ \forall\,(T,A) \in \mathcal{S}\ :$

$$Tx \in \mathrm{supp}\,(\mu) \qquad \text{and} \qquad A_x \bar{\nu}_x \,=\, \bar{\nu}_{Tx}.$$

*Proof.* We begin by verifying that:

$$\forall\,(T,A) \in \mathcal{S}, \quad \forall\, x \in \mathrm{supp}\,\mu \ : \quad Tx \in \mathrm{supp}\,\mu.$$

1. From the Lemma 36 and since $\mathrm{supp}\,(\mu)$ has full $\mu$-measure, we have from stationarity that
$$\forall\, t \in \mathrm{T}, \quad \mu\text{-}\forall\exists\, x \in \mathrm{X} \ : \quad \mathrm{P}_t(x,\, \mathrm{supp}\,(\mu)) \,=\, 1.$$
2. For strongly continuous RDS $\mathcal{T}$ with independent increments, the mapping $x \longmapsto \mathrm{P}_t(x,\cdot)$ is weak-$*$ continuous. Thus, by the Portmanteau Theorem[2] and the density of $\mu$-almost surely sets in $\mathrm{supp}\,(\mu)$, we conclude that:
$$\forall\, t \in \mathrm{T}, \quad \forall\, x \in \mathrm{supp}\,(\mu) \ : \quad \mathrm{P}_t(x,\, \mathrm{supp}\,(\mu)) \,=\, 1.$$

So, for any fixed $x \in \mathrm{X}$, we have for all $t \in \mathrm{T}$ that:

$$\mathcal{T}^t_\omega x \in \mathrm{supp}\,(\mu) \ \text{ with probability 1.}$$

In particular, any $(T,A) \in \mathcal{S}_t$ is the limit (in the topology on $\mathcal{C}$) of elements $(T^n, A^n) \in \mathcal{S}_t$ for which $T^n x \in \mathrm{supp}\,(\mu)$ for all $n$. Therefore, by the closedness of $\mathrm{supp}\,(\mu)$ for any $(T,A) \in \mathcal{S}_t$, we have: $Tx \in \mathrm{supp}\,(\mu)$. A similar argument implies that for any $(T,A) \in \mathcal{S}\ :\ Tx \in \mathrm{supp}\,(\mu)$.

Let us now move on to verifying the relation. We fix $x \in \mathrm{supp}\,(\mu)$ and $(T,A) \in \mathcal{S}$, and define

$$G_x \,=\, \{(T,A) \in \mathcal{C}\ :\ (A_x)_* \bar{\nu}_x \,=\, \bar{\nu}_{Tx}\}\,.$$

Note that by the argument in the previous paragraph, $\bar{\nu}_{Tx}$ is defined for all $x \in \mathrm{supp}\,(\mu)$ and $(T,A) \in \mathcal{S}$. To complete the proof of the proposition, it suffices to show that:

$$\forall\, x \in \mathrm{supp}\,(\mu) \ : \quad \mathcal{S} \subset G_x.$$

By the previous lemma, we have the existence of $\widetilde{\mathrm{X}}$.

1. By the closedness of $\mathrm{supp}\,(\mu)$ and the fact that $x \longmapsto \bar{\nu}_x$ is weak-$*$ continuous, we have that: $G_x$ is closed in $\mathcal{C}$. Hence:
$$\forall\, x \in \widetilde{\mathrm{X}}, \quad \forall\, t \in \mathrm{T} \ : \quad \mathbb{P}\left(\left\{\omega \in \Omega \ :\ (\mathcal{T}^t_\omega, \mathcal{A}^t_{\omega,(\cdot)}) \in G_x\right\}\right) \,=\, 1,$$
and then $G_x$ is dense in $\mathcal{S}_t$.
2. Since $G_x$ and $\mathcal{S}_t$ are closed in $\mathcal{C}$, we have that: $\mathcal{S}_t \subset G_x$.
3. Moreover, for all $t \in \mathrm{T}$, there exists a sequence $(t_n)$ for which $t_n \underset{n+\infty}{\longrightarrow} t$ and such that $(T,A) \in \mathcal{S}_t$ is a limit of elements $(T^n, A^n) \in \mathcal{S}_{t_n}$ in $\mathcal{C}$.
4. Again by closedness of $G_x$ we deduce that
$$\forall\, t \in \mathrm{T} \ : \quad \mathcal{S}_t \subset G_x,$$
and conclude that $\mathcal{S} \subset G_x$ for all $x \in \widetilde{\mathrm{X}}$.

To conclude for $x \in \mathrm{supp}\,(\mu) \backslash \widetilde{\mathrm{X}}$, we consider a convergent sequence, $x_n \longrightarrow x$ where $x_n \in \widetilde{\mathrm{X}}$, and fix $(T,A) \in \mathcal{S}$. Since $(T,A) \in G_{x_n}$ for all $n$ from above and from the continuity of $x \longmapsto \bar{\nu}_x$, we have that $(T,A) \in G_x$. □

[2] Portmanteau Theorem For $(\mu_n), \mu \in \mathcal{P}(\mathrm{X})$ we have:

$$\mu_n \rightharpoonup \mu \qquad \Longleftrightarrow \qquad \forall\, A \in \mathcal{B}_{\mathrm{X}},\ \mu(\partial A) = 0 \ : \quad \lim_{n+\infty} \mu_n(A) = \mu(A).$$

**Statement of the result**

**Theorem 40** (Classification of invariant fiber measure families [3])**.** *Let $d \leqslant 3$ and $(\bar{\nu}_x)_{x\in\operatorname{supp}(\mu)}$ be the invariant measure family so-obtained. Then, one of the following alternatives holds.*

*(a) For all $x \in \operatorname{supp}(\mu)$, there exists an inner product $\langle\cdot,\cdot\rangle_x$ on $\mathbf{R}^d$ such that: $x \longmapsto \langle\cdot,\cdot\rangle_x$ is continuous and,*

$$\forall\,(T,A)\in\mathcal{S},\quad \forall\,x\in\operatorname{supp}(\mu)\ :\quad A_x : (\mathbf{R}^d,\langle\cdot,\cdot\rangle_x)\longrightarrow(\mathbf{R}^d,\langle\cdot,\cdot\rangle_{Tx}) \text{ is an isometry.}$$

*(b) There exists $p \geqslant 1$, such that for all $x \in \operatorname{supp}(\mu)$, there exist proper, distinct, nontrivial linear subspaces $\mathrm{E}^1_x,\dots,\mathrm{E}^p_x \subsetneq \mathbf{R}^d$ such that $x \longmapsto \mathrm{E}^i_x$ is measurable and:*

*(i)* $\bar{\nu}_x\left(\bigcup_{i=1}^{p}\mathrm{E}^i_x\right) = 1,$

*(ii)* $\forall\,(T,A)\in\mathcal{S}\ :\quad A_x\mathrm{E}^i_x = \mathrm{E}^{\pi(i)}_{Tx},\quad$ *where* $\pi = \pi_{(T,A)} \in \mathfrak{S}_p,$

*(iii) the collection $(\mathrm{E}^i_x)$ is locally continuous up to re-labelling: there is an open neighborhood $\mathrm{U}\subset\mathrm{X}$ of $x$ and a labelling of the subspaces $\mathrm{E}^i_x, x\in\mathrm{U}\cap\operatorname{supp}(\mu)$ such that*

$$\begin{array}{ccc}\mathrm{U}\cap\operatorname{supp}(\mu) & \longrightarrow & \mathcal{P}(\mathbf{R}^d)\\ y & \longmapsto & \mathrm{E}^i_y\end{array}\quad\text{is continuously varying}$$

*Remark.* The restriction to $d \leqslant 3$ is only relevant in case (b). For $d \geqslant 4$ the result is likely to be true, but the proof is lengthier due to the fact that among the $\mathrm{E}^i_x, x\in\operatorname{supp}(\mu), 1\leqslant i\leqslant p$ there may be arbitrarily many subspaces of dimension $\geqslant 2$.

**Proof of the classification**

The germ of this idea comes from the geometry of $\mathrm{SL}_d(\mathbf{R})$ and the restrictions placed on the subgroup of matrices preserving a single projective measure:

**Lemma 41** (Furstenberg [15])**.** *Let $d \geqslant 1$, $\eta$ be a Borel measure on $\mathrm{P}^{d-1}$ and:*

$$\mathrm{H} := \mathrm{H}_\eta := \{A\in\mathrm{SL}_d(\mathbf{R})\ :\ A_*\eta = \eta\}.$$

*Then,* H *is closed and:*

- *if* H *is compact, then there is an inner product $\langle\cdot,\cdot\rangle'$ on $\mathbf{R}^d$ (with corresponding norm $\|\cdot\|'$) such that*

$$\text{every } A\in\mathrm{H} \text{ is an isometry.}$$

- *if* H *is non-compact, then there exist distinct, proper, nontrivial linear subspaces $\mathrm{E}^1,\dots,\mathrm{E}^p\subset\mathbf{R}^d$, such that:*

*(i)* $\eta\left(\bigcup_i \mathrm{E}^i\right) = 1,$

*(ii)* $\forall\,A\in\mathrm{H},\quad\exists\,\pi\in\mathfrak{S}_p\ :\quad\forall\,i\in[\![1,p]\!],\quad A\mathrm{E}^i = \mathrm{E}^{\pi(i)},$

*(iii) $\forall\,i\in[\![1,p]\!]$, there is an inner product $\langle\cdot,\cdot\rangle^i$ on $\mathrm{E}^i$ such that:*

$$\forall\,A\in\mathrm{H},\quad A_{|\mathrm{E}^i} \text{ is conformal with respect to the inner products } \langle\cdot,\cdot\rangle^i, \langle\cdot,\cdot\rangle^{\pi(i)} \text{ respectively.}$$

*Proof.* This lemma is proved in the appendix (section B.2). □

We can now prove the classification theorem.

*Proof of the classification.* For $x \in \operatorname{supp}(\mu)$, define

$$O_x \,=\, \{Tx \,:\, (T,A) \in \mathfrak{S}_p\}\,.$$

Note that $O_x \subset \operatorname{supp}(\mu)$ holds by a previous lemma. Using ergodicity of $\mu$ and the strong Feller property, we get the following preliminary Lemma.

**Lemma 42.**

$$\forall\, x \in \operatorname{supp}(\mu) \,:\quad \mu(O_x) \,=\, 1.$$

*Proof of Lemma 42.* Let $x \in \operatorname{supp}(\mu)$.

1. Fix $t > 0$ and let $y \in O_x$, then there exists $(T,A) \in \mathfrak{S}_p \,:\, y = Tx$. Now, fix a $\mathbb{P}$-generic $\omega \in \Omega$ and set $T' = \mathcal{T}^t_\omega$ and $A' = \mathcal{A}^t_{\omega,(\cdot)}$.
2. Noting $(T',A') \in \mathcal{S}$ with probability 1, we see that $T'y = T' \circ Tx$, hence $T'y \in O_x$.
3. Since $y \in O_x$ was arbitrary, we conclude $\mathcal{T}^t_\omega y \in O_x$ for any $t \geqslant 0$ with probability 1: hence $O_x$ is $(\mathrm{P}_t, \mu)$-invariant.
4. It follows from ergodicity for $\mu$ that $O_x$ has zero or full $\mu$-measure. Assume that by stationarity, $P_t(y, O_x) = 0$ for $\mu$-almost all $y \in Z$.
5. From the ultra Feller property for the semigroup $(\mathrm{P}_t)$, we conclude $P_t(x, O_x) = 0$, a contradiction because $\{y \in Z : P_t(y, O_x) = 0\}$ must be dense in $\operatorname{supp}(\mu)$. We conclude $\mu(O_x) > 0$, hence $\mu(O_x) = 1$.

□

Fix $x_0 \in \operatorname{supp}(\mu)$ and consider the following $\mathrm{SL}_2(\mathbf{R})$ subgroup:

$$\mathrm{H}_{x_0} \,=\, \{A \in \mathrm{SL}_2(\mathbf{R}) : A_*\bar{\nu}_{x_0} \,=\, \bar{\nu}_{x_0}\}\,.$$

Note that $\mathrm{H}_{x_0}$ is closed by Lemma 41. There are two cases:

(a) **Assume that $\mathrm{H}_{x_0}$ is compact.**

By Lemma 41 there is an inner product $\langle\cdot,\cdot\rangle_{x_0}$ with respect to which all matrices in $\mathrm{H}_{x_0}$ act as isometries.

For each $x \in \operatorname{supp}(\mu)$ we have that $\mu(O_{x_0} \cap O_x) = 1$ by Lemma 60, and then, there exists a point $y \in O_{x_0} \cap O_x$. Let $(T,A),(S,B) \in \mathcal{S}$ be such that:

$$Tx_0 \,=\, y \qquad \text{and} \qquad Sx \,=\, y.$$

We hence define the inner product $\langle\cdot,\cdot\rangle_x$ for all $u, v \in \mathbf{R}^d$ by:

$$\langle u, v\rangle_x \,=\, \left\langle A_{x_0}^{-1} \circ B_x u,\, A_{x_0}^{-1} \circ B_x v\right\rangle_{x_0}.$$

This definition does not depend on the exact choice of $(T,A),(S,B)$. Indeed, if $(\bar{T},\bar{A}),(\bar{S},\bar{B}) \in \mathcal{S}$ are any other elements for which

$$\bar{T}x_0 \,=\, y \qquad \text{and} \qquad \bar{S}x \,=\, y,$$

then Proposition 39 implies $(\bar{A}_{x_0})^{-1}\bar{B}_x(B_x)^{-1}A_{x_0} \in \mathrm{H}_{x_0}$, and so by Lemma 41 (a):

$$\left\langle A_{x_0}^{-1} \circ B_x u,\, A_{x_0}^{-1} \circ B_x v\right\rangle_{x_0} \,=\, \left\langle \bar{A}_{x_0}^{-1} \circ \bar{B}_x u,\, \bar{A}_{x_0}^{-1} \circ \bar{B}_x v\right\rangle_{x_0}.$$

By a similar proof, one checks that for each $(T,A) \in \mathcal{S}$ and $x \in \operatorname{supp}(\mu)$, we have that:

$$A_x : (\mathbf{R}^d, \langle\cdot,\cdot\rangle_x) \longrightarrow (\mathbf{R}^d, \langle\cdot,\cdot\rangle_{Tx}) \text{ is an isometry.}$$

For each $x \in \operatorname{supp}(\mu)$, the inner product $\langle \cdot, \cdot \rangle_x$ gives rise to a Euclidean volume on $\mathbf{R}^d$ and an induced volume $\widetilde{\nu}_x$ on $\mathrm{P}^{d-1}$. By the isometry property, it follows that:

$$\forall\, (T, A) \in \mathcal{S}, \quad \forall\, x \in \operatorname{supp}(\mu) \ : \quad (A_x)_* \widetilde{\nu}_x \ = \ \widetilde{\nu}_{Tx}.$$

Thus $(\widetilde{\nu}_x)_{x \in \operatorname{supp}(\mu)}$ defines an invariant measure family on $\operatorname{supp}(\mu)$. Repeating the proof of Proposition 37 for this new invariant measure family, we conclude that $x \longmapsto \widetilde{\nu}_x$ is weak-$*$ continuous. Hence, from the fact that for all $x$ : $\widetilde{\nu}_x \ll \mathrm{Leb}_{\mathrm{P}^{d-1}} =: \lambda$, we conclude that the densities:

$$x \ \longmapsto \ \rho_x \ := \ \frac{d\widetilde{\nu}_x}{\mathrm{d}\lambda},$$

vary continuously in the uniform norm on $C(\mathrm{P}^{d-1}, \mathbf{R})$. It is now straightforward to check that the corresponding inner products $x \longmapsto \langle \cdot, \cdot \rangle_x$ is continuous.

(b) **Now assume that $\mathrm{H}_{x_0}$ is noncompact.**

Let $\mathrm{E}^i_{x_0} = \mathrm{E}^i, 1 \leqslant i \leqslant p$ be as in Lemma 41 $(b)$ applied to $\mathrm{H} = \mathrm{H}_{x_0}$. For each $i$, let $\langle \cdot, \cdot \rangle^i$ denote the corresponding inner product on $\mathrm{E}^i = \mathrm{E}^i_{x_0}$.

Let $x \in \operatorname{supp}(\mu)$, there exists $y \in O_{x_0} \cap O_x$, as in the proof for case (a), and fix $(T, A), (S, B)$ for which $Tx_0 = y, Sx = y$. We then define:

$$\mathrm{E}^i_x \ := \ (B_x)^{-1} \circ A_{x_0}(\mathrm{E}^i_{x_0}).$$

We also define the inner products $\langle \cdot, \cdot \rangle^i_x$ on $\mathrm{E}^i_x$ by setting, for $v, w \in \mathrm{E}^i_x$,

$$\langle v, w \rangle^i_x \ = \ \left\langle (A_{x_0})^{-1} B_x v, \, (A_{x_0})^{-1} B_x w \right\rangle^i.$$

As in the proof of case (a), one checks that the above definitions do not depend on the exact choices of $y \in O_x \cap O_{x_0}$ or $(T, A), (S, B) \in \mathcal{S}$. By a similar check, the invariance property for the $\mathrm{E}^i_x, x \in \operatorname{supp}(\mu)$ similarly holds, and moreover, for $(T, A) \in \mathcal{S}$ and $x \in \operatorname{supp}(\mu)$, we have that:

$$A_x : (\mathrm{E}^i_x, \langle \cdot, \cdot \rangle^i_x) \longrightarrow (\mathrm{E}^{\pi(i)}_{Tx}, \langle \cdot, \cdot \rangle^{\pi(i)}_{Tx}) \ \text{ is conformal.}$$

Now observe that since $d \leqslant 3$, there are two cases:

$$\forall\, i \ : \quad \dim \mathrm{E}^i_x \equiv 1 \qquad \text{or} \qquad \text{for some } i, x \ : \quad \dim \mathrm{E}^i_x \ = \ 2.$$

In the first case, since $(\bar{\nu}_x)_{|\mathrm{E}^i_x}$ is a delta mass supported on the projective point corresponding to $\mathrm{E}^i_x$, hence the local continuity of $x \longmapsto \mathrm{E}^i_x$ up to relabeling follows immediately.

In the second case, by Lemma 58 we must have that $p \leqslant 2$ and that at most one of the $\mathrm{E}^i_x$ is twodimensional for each $x \in \operatorname{supp}(\mu)$. We focus on the case $p = 1$ (essentially the same proof applies when $p = 2$). Hereafter let us write $\mathrm{E}_x \ := \ \mathrm{E}^1_x$. Note that this can only occur when $d = 3$, which hereafter we assume.

In analogy with the proof of Theorem 40 (a), consider for each $x \in \operatorname{supp}(\mu)$, the Euclidean volume $m_x$ on $\mathrm{E}_x \subset \mathbf{R}^3$ induced by the inner product $\langle \cdot, \cdot \rangle_x \ := \ \langle \cdot, \cdot \rangle^1_x$. This induces a normalized volume $\widetilde{\nu}_x$ on the projectivization of $\mathrm{E}_x$ in $\mathrm{P}^2$. As in the proof for case (a), the fiber measure family $(\widetilde{\nu}_x)_{x \in \operatorname{supp}(\mu)}$ is invariant. This follows from the conformality property for the inner product $\langle \cdot, \cdot \rangle_x$. As in the proof of case (a), we can repeat the arguments of Proposition 37, from which we obtain that the family $(\widetilde{\nu}_x)$ is weak-$*$ continuous. Continuity of $x \longmapsto \mathrm{E}_x$ now follows. $\square$

**Approximate controllability criteria**

Finally, we state a weakening ruling out each alternative in the classification we obtain. Let see a sufficient condition for $\lambda_1 > 0$ : approximate controllability criteria.

**Definition 37.** We say that the cocycle $\mathcal{A}$ has an **approximate controllability** if there exists $x, y \in \operatorname{supp}(\mu)$ such that:

$(i)$ $\exists\, t_0 > 0 \ : \ y \in \operatorname{supp}(\mathrm{P}_{t_0}(x, \cdot))$,

$(ii)$ $\forall\, \varepsilon, \mathrm{M} > 0 \ : \ \mathrm{Q}_{t_0}((x, \mathrm{id}), \mathrm{B}_\varepsilon(y) \times \{A \in \mathrm{SL}_d(\mathbf{R}) : |A| > M\}) > 0$,

$(iii)$ $\forall\, v \in \mathrm{P}^{d-1}$, $\forall\, \mathrm{V} \subset \mathrm{P}^{d-1}$ open, $\forall\, \varepsilon > 0$:

$$\widehat{\mathrm{P}}_{t_0}((x, v), \mathrm{B}_\varepsilon(y) \times \mathrm{V}) > 0.$$

We can now prove the following.

**Theorem 43** (Sufficient condition to rule out the criterion)**.** *Let $d \leqslant 3$, $\mathcal{T}$ be a strongly continuous RDS with independent increments and $\mathcal{A}$ be an $\mathrm{SL}_d(\mathbf{R})$ invertible, integrable linear cocycle over $\mathcal{T}$ such that the associated projective RDS $\mathcal{P}$ has projective independent increments. If:*

- *$(\mathrm{P}_t)$ is strong Feller,*
- *$\mathcal{T}$ admits an ergodic stationary measure $\mu$ for which $\mathcal{A}$ has an approximate controllability,*

*then:*

$$\lambda^+ > \lambda^-.$$

**Corollary 44.** *In particular $\lambda_1 > 0$.*

*Proof.* Assume that $\lambda^+ = \lambda^-$. Then the Furstenberg's criterion applies, and so either case $(a)$ for case $(b)$ holds in the classification of the invariant fiber measure family $(\nu_x)$. For $y \in \operatorname{supp}\mu$, we write $|\cdot|_y$ for the norm corresponding to the inner product $\langle \cdot, \cdot \rangle_y$.

Let's start by ruling out $(a)$.

(1) Let $\kappa = \max\left( \max_{v \in \mathbf{R}^d \setminus \{0\}} \dfrac{|v|_x}{|v|}, \max_{v \in \mathbf{R}^d \setminus \{0\}} \dfrac{|v|_y}{|v|} \right)$, and fix $\varepsilon > 0$ so that:

$$\forall\, x \in \mathrm{B}_\varepsilon(y) \ : \quad \frac{1}{2}\,|\cdot|_x \leqslant |\cdot|_y \leqslant 2\,|\cdot|_x\,.$$

(2) From the point $(ii)$ of the definition 37, there is a $\mathbb{P}$-positive measure set $\mathrm{E} \subset \Omega$ such that:

$$\forall\, \omega \in \mathrm{E} \ : \quad \mathcal{T}^{t_0}_\omega x \in \mathrm{B}_\varepsilon(y) \qquad \text{and} \qquad \left\| \mathcal{A}^{t_0}_{\omega,x} \right\| > 2\kappa^2.$$

Without loss we can assume $\left\{ (\mathcal{T}^{t_0}_\omega, \mathcal{A}^{t_0}_{\omega,\cdot}) : \omega \in \mathrm{E} \right\} \subset \mathcal{S}_{t_0}$, perhaps on paring off an $\mathbb{P}$-measure zero set from E.

(3) By (a):

$$\forall\, \omega \in \mathrm{E} \ : \quad \left\| \mathcal{A}^{t_0}_{\omega,x} \right\|_{x,y} = 1,$$

where $|\cdot|_{x,y}$ is the matrix norm induced by the norms $|\cdot|_x$ at $x$ and $|\cdot|_y$ at $y = \mathcal{T}^{t_0}_\omega x$.

(4) From this we obtain the estimate $\left\| \mathcal{A}^{t_0}_{\omega,x} \right\| \leqslant 2\kappa^2$ in the matrix norm induced from $|\cdot|$, which is a contradiction to $(ii)$.

Turning to case (b).

(1) Take $\varepsilon > 0$ sufficiently small so that:
   - a labelling of the $E^i_y, y \in \mathrm{B}_\varepsilon(y)$ exists for which $y \mapsto \mathrm{E}^i_y$ is continuous for $1 \leqslant i \leqslant p$,
   - there is an open set $V \subset \mathrm{P}^{d-1}$ for which $V \cap \left(\cup_i \mathrm{E}^i_y\right) = \emptyset$ for all $y \in \mathrm{B}_\varepsilon(y)$,

   and fix an arbitrary $1 \leqslant i \leqslant p$ and $v \in \mathrm{E}^i_x \setminus \left(\cup_{j \neq i} \mathrm{E}^j_x\right)$.

(2) Condition $(iii)$ implies that there is a $\mathbf{P}$-positive measure set $E \subset \Omega$ such that for all $\omega \in \mathrm{E}$, we have $\mathcal{T}^{t_0}_\omega x \in \mathrm{B}_\varepsilon(y)$ and $\mathcal{A}^{t_0}_{\omega,x} v \in V$.

(3) As before, on paring off a $\mathbf{P}$-measure xero set we can assume $\left(\mathcal{T}^{t_0}_\omega, \mathcal{A}^{t_0}_{\omega,(\cdot)}\right) \in \mathcal{S}_{t_0}$ for all $\omega \in \mathrm{E}$, from which we deduce from the classification that $\mathcal{A}^{t_0}_{\omega,x} \mathrm{E}^j_x = E^{\pi_\omega(j)}_{\mathcal{T}^{t_0}_\omega x}$ for all $\omega \in \mathrm{E}$ and $1 \leqslant j \leqslant p$, where $\pi_\omega$ is some permutation on $\{1, \cdots, p\}$.

(4) But at $j = i$ this is a contradiction, since $v \in \mathrm{E}^i_x$ yet $\mathcal{A}^{t_0}_{\omega,x} v \notin \mathrm{E}^l_{\mathcal{T}^{t_0}_\omega x}$ for any $l \in \{1, \cdots, p\}$ by construction.

□

## 5. Lagrangian chaos

### 5.1. Introduction

We study here the property of chaos (in terms of sensitivity with respect to initial conditions) of a turbulent fluid. Hence, we want to prove that in many circumstances the Lagrangian flow associated to velocity fields are chaotic. To this end, we focus on what happens when particles move together initially: theses two points follow chaotic paths and go off in their own directions.

#### Statement of the main result

To calculate the *mean time* before two points of the fluid diverge exponentially, we introduce a precious tool which characterise this property of chaos: the *Lyapunov exponent*.

Recall that we consider $(u_t)$ the vector field process on $\mathbf{H}$ governed by Systems 1-2. Thus, we study the stochastic flow of diffeomorphisms $\phi^t : \mathbb{T}^d \longrightarrow \mathbb{T}^d$, $t \geqslant 0$ defined by the random ODE:

$$\begin{cases} \dfrac{\mathrm{d}\phi^t}{\mathrm{d}t}(x) = u_t(\phi^t(x)), & t > 0, \\ \phi^0(x) = x & \in \mathbb{T}^d. \end{cases}$$

This map is random because $(u_t)$ is random. We then consider the one-point particle motion $(x_t)$ for all $t \geqslant 0$ and $x \in \mathbb{T}^d$ by:

$$x_t = \phi^t(x).$$

Note that $\phi_t(x)$ is not a Markov process. In addition, we consider the following Lagrangian process on $\mathbf{H} \times \mathbb{T}^d$:

$$\begin{cases} z_t = (u_t, x_t), & \forall\, t > 0, \\ z_0 = (u, x) & \in \mathbf{H} \times \mathbb{T}^d. \end{cases}$$

By the regularity of the velocity fields, the process $(z_t)$ is as well-posed as $(u_t)$.

**Theorem 45** (Positive Lyapunov exponent [3])**.** *Let $(u_t)$ be governed by the previous system. Then, for all $\nu > 0$, there exists a deterministic constant $\lambda^+ > 0$ such that for every initial vector field $u_0 \in \operatorname{supp} \mu$ and $x \in \mathbb{T}^d$, the following limit exists p.s :*

$$\lambda^+ = \lim_{t+\infty} \frac{1}{t} \ln \left| D_x \phi^t \right| > 0 \qquad \text{a.s.}$$

*Remark.* In particular, every particule in every experiment finally got the same exponent sensibility, and vortices cannot permanently trap any particules.

### 5.2. **Proof of the Lagrangian choas**

*Outline of the proof.* The proof show that, given appropriate ergodic properties of the dynamics, the Lyapunov exponent is strictly positive *unless* there is a certain p.s *invariant certain structure* in the motion of $x_t$ and its gradient $D_{x_0}\phi^t$.

1. First, we use the MET to prove that the limit defining the Lyapunov exponent $\lambda^+$ exists and is constant almost surely.
2. Then, we exclude the invariant structure and deduce the positivity of the Lyapunov exponent by using the classification of fiber measures to prove that this exponent satisfies $\lambda^+ > 0$. Moreover, we need hypoellipticity and approximate controllability arguments to show that: $(i)$ dynamics shows suitable ergodic properties, $(ii)$ many moves of $x_t$ and $D_{x_0}\phi^t$ are realized.

**RDS framework**

Given a random noise path $\omega \in \Omega$, a time $t$ and a fixed initial $(u, x) \in \mathbf{H} \times \mathbb{T}^d$, we denote the mapping send $(\omega, t, u)$ to the time-$t$ vector field $u_t$ conditioned on $u_0 = u$:

$$\begin{array}{rccc} \mathcal{U}_\omega : & \mathrm{T} \times \mathbf{H} & \longrightarrow & \mathbf{H}, \\ & (t, u) & \longmapsto & u_t. \end{array}$$

We now introduce a random noise path for our one-point motion by an auxiliary process:

$$x_t = \phi^t_{\omega,u} x,$$

which is well-posed. The corresponding mapping for the Lagrangian flow process $(u_t, x_t)$ is:

$$\begin{array}{rccc} \mathcal{T}_\omega : & \mathrm{T} \times \mathbf{H} \times \mathbb{T}^d & \longrightarrow & \mathbf{H} \times \mathbb{T}^d, \\ & (t, u, x) & \longmapsto & (\mathcal{U}^t_\omega u,\, \phi^t_{\omega,u} x). \end{array}$$

Consider $(v_t)$, $(\check{v}_t)$ and $(A_t)$ which are defined respectively on $\mathrm{P}^{d-1}$ and $\mathcal{M}_d(\mathbf{R})$ by:

$$v_t := D_x\phi^t_{\omega,x} v, \qquad \check{v}_t := (D_x\phi^t_{\omega,u})^{-\top}\check{v} \qquad \text{and} \qquad A_t := D_x\phi^t_{\omega,u},$$

and are generated by:

$$\partial_t v_t = \Pi_{v_t}\nabla u(x_t)v_t, \qquad \partial_t \check{v}_t = -\Pi_{\check{v}_t}\nabla u_t(x_t)^\top \check{v}_t \qquad \text{and} \qquad \partial A_t = \nabla u_t(x_t)A_t,$$

where $\Pi_v = \mathrm{id} - v \otimes v$ is the orthogonal projection from $\mathbf{R}^d$ onto the tangent space of $\mathbb{S}^{d-1}$. We write $(\mathcal{P}_\omega)$ and $(\check{\mathcal{P}}_\omega)$ for the projective processes on $\mathbf{H} \times \mathbb{T}^d \times \mathrm{P}^{d-1}$ of $(u_t, x_t, v_t)$ and $(u_t, x_t, \check{v}_t)$ respectively, and $(\mathcal{M}_\omega)$ for the projective process on $\mathbf{H} \times \mathbb{T}^d \times \mathrm{SL}_d(\mathbf{R})$ of $(u_t, x_t, A_t)$. In addition, we consider the linear cocycles defined for $(\omega, t, u, x) \in (\Omega, \mathrm{T}, \mathbf{H}, \mathbb{T}^d)$ by:

$$\mathcal{A}^t_{\omega,u,x} := D_x\phi^t_{\omega,u},$$

and $\check{\mathcal{A}}^t_{\omega,u,x} := (\mathcal{A}^t_{\omega,u,x})^{-\top}$.

**Proposition 46** (Basic properties)**.**

$(i)$ *$(\mathcal{U}_\omega)$, $(\mathcal{T}_\omega)$, $(\mathcal{P}_\omega)$, $(\check{\mathcal{P}}_\omega)$ and $(\mathcal{M}_\omega)$ are strongly continuous RDS with independent increments.*

$(ii)$ *$\mathcal{A}$ and $\check{\mathcal{A}}$ are invertible, integrable and has independent increments.*

*Proof.*

$(i)$ These properties follow from Proposition 2.

$(ii)$ The integrability condition for each of these processes follows from Proposition 2, while the independent increments condition is equivalent to that of $(\mathcal{M}_\omega)$.

□

**Hypoellipticity: uniqueness of the stationary measure & strong Feller**

There have been a number of works proving uniqueness of the stationary measure of the Navier-Stokes equations under degenerate noise. A standard approach is to apply the Khasminskii Theorem: since distinct ergodic stationary measures for strong Feller processes have disjoint supports, we check that there exists a point which belongs to the support of every invariant measure (*i.e.*, weak irreducibility).

**Proposition 47** (Strong Feller)**.** *For Systems 1-2, the Markov semigroups associated with* $(\mathcal{T}_\omega)$*,* $(\mathcal{P}_\omega)$ *and* $(\check{\mathcal{P}}_\omega)$ *are strong Feller in* $\mathbf{H} \times \mathbb{T}^d \times \mathrm{P}^{d-1}$*.*

*Proof.* Proved in section 6 of [3]. □

To conclude uniqueness of the stationary measures it suffices to prove the following weak irreducibility properties.

**Proposition 48** (Weak irreducibility properties)**.** *For systems 1-2 we have the following.*

*(1) The support of any stationary measure for* $(\mathcal{T}_\omega)$ *on* $\mathbf{H} \times \mathbb{T}^d$ *must contain the set* $\{0\} \times \mathbb{T}^d$*.*

*(2) The support of any stationary measure for* $(\mathcal{P}_\omega)$ *and* $(\check{\mathcal{P}}_\omega)$ *on* $\mathbf{H} \times \mathbb{T}^d \times \mathrm{P}^{d-1}$ *must contain* $\{0\} \times \mathbb{T}^d \times \mathrm{P}^{d-1}$*.*

*Proof.* Proved in section 7 of [3]. □

**Corollary 49.** *The processes* $(\mathcal{T}_\omega)$*,* $(\mathcal{P}_\omega)$ *and* $(\check{\mathcal{P}}_\omega)$ *all have unique stationary measures.*

**Approximate controllability: Furstenberg's criterion for infinite-dimensional models**

For our infinite-dimensional models, we are not aware of any means by which one can prove a positive density for the conditional law of $D_x\phi^t_{\omega,u}$. Instead, we are able to prove a certain *approximate controllability* statement. To articulate this we use the projective process $(u_t, x_t, v_t)$ on $\mathbf{H} \times \mathbb{T}^d \times \mathrm{P}^{d-1}$.

**Proposition 50** (Approximate controllability condition)**.** *For any* $x, y \in \mathbb{T}^d$ *and* $t > 0$*, we have the following:*

*(a) for any* $\varepsilon, M > 0$*, we have that*

$$\mathbb{P}\left((u_t, x_t) \in \mathcal{B}_\varepsilon(0) \times \mathcal{B}_\varepsilon(y),\ |A_t| > M \mid u_0 = 0,\ x_0 = x,\ A_0 = \mathrm{id}\right) > 0.$$

*(b) for any* $\varepsilon > 0, v \in \mathrm{P}^{d-1}$ *and open* $V \subset \mathrm{P}^{d-1}$*, we have*

$$\mathbb{P}\left((u_t, x_t) \in \mathcal{B}_\varepsilon(0) \times \mathcal{B}_\varepsilon(y),\ v_t \in V \mid u_0 = 0,\ x_0 = x,\ v_0 = v\right) > 0.$$

*Proof.* Proved in [3]. □

**End of the proof**

Since $(\mathcal{T}_\omega)$, $(\mathcal{P}_\omega)$ and $(\check{\mathcal{P}}_\omega)$ have unique stationary measures we then deduce respectively: the almost-sure constancy of $\lambda^+$ and $\check{\lambda}^+$, the almost-sure growth for $(D_x\phi^t)v$, and the growth of $(D_x\phi^t)^{-\top}v$.

Then, the Strong Feller property proved for $(\mathcal{T}_\omega)$ allows us to justify the refinement of Furstenberg's criterion.

Finally, from the approximate controllability condition we can rule out the refinement of Furstenberg's criterion for our system, and then complete the proof of Lagrangian chaos.

## A. **Markov processes**

Let:

- $(\mathrm{T}, +)$ be a semi-group,
- $(\Omega, \mathcal{F})$ be a measurable space with a filtration $\{\mathcal{F}_t, t \in \mathrm{T}\}$,
- X be a Polish space,
- $(\mathbb{P}_x)_{x\in\mathrm{X}}$ a family of probability measures on $(\Omega, \mathcal{F})$ such that:

$$\forall\, \mathrm{A} \in \mathcal{F} \ : \quad x \longmapsto \mathbb{P}_x(\mathrm{A}) \ \text{ is universally measurable.}$$

**Definition 38.** A **stochastic process** $(x_t)_{t\in\mathrm{T}}$ is a family of X-valued random variables $x_t$ on $(\Omega, \mathcal{F})$ indexed by a set T.

*Remark.* For $\omega \in \Omega$ fixed, the mapping $t \longmapsto x_t(\omega)$ is a real valued function called the process **path**.

**Definition 39.** For each stochastic process $(x_t)_{t\in\mathrm{T}}$ we define the **transition function** as the law of $x_t$ under the probability measure $\mathbb{P}_x$:

$$\forall\, x \in \mathrm{X}, \quad \forall\, \mathrm{K} \in \mathcal{B}_\mathrm{X} \quad : \quad \mathrm{P}_t(x, \mathrm{K}) \ := \ \mathbb{P}_x(x_t \in \mathrm{K}).$$

**Definition 40.** A **Markov process** in X is a stochastic process $(x_t)_{t\in\mathrm{T}}$ adapted to the filtration $\mathcal{F}_t$ and satisfying:

$$\forall\, x \in \mathrm{X}, \quad \forall\, \mathrm{K} \in \mathcal{B}_\mathrm{X}, \quad \forall\, t, s \in \mathrm{T} :$$
$$\mathbb{P}_x(x_0 \,=\, x) \,=\, 1 \qquad \text{and} \qquad \mathbb{P}_x\text{-}\forall\!\exists\, \omega \in \Omega \ : \quad \mathbb{P}_x(x_{t+s} \in \mathrm{K} \mid \mathcal{F}_s) \,=\, \mathrm{P}_t(x_s, \mathrm{K}).$$

*Remark.* We denote "for $\mu$-almost every" as "$\mu$-$\forall\!\exists$".

**Definition 41.** Given $\lambda \in \mathcal{P}(\mathrm{X})$ and a Markov process $(x_t, \mathbb{P}_x)$, we define the probability measure

$$\begin{array}{rccc} \mathbb{P}_\lambda : & \mathcal{F} & \longrightarrow & \mathbf{R} \\ & \mathrm{A} & \longmapsto & \displaystyle\int_\mathrm{X} \mathbb{P}_x(\mathrm{A})\lambda(\mathrm{d}x), \end{array}$$

and denote by $\mathbb{E}_\lambda$ the corresponding mean value.

**Lemma 51** (Kolmogorov-Chapman)**.**

$$\forall\, x \in \mathrm{X}, \quad \forall\, \mathrm{K} \in \mathcal{B}_\mathrm{X}, \quad \forall\, t, s \in \mathrm{T} :$$
$$\mathrm{P}_{t+s}(x, \mathrm{K}) \,=\, \int_\mathrm{X} \mathrm{P}_t(x, \mathrm{d}y)\mathrm{P}_s(y, \mathrm{K}).$$

*Remark.* This relation implies that every Markov process generates an evolution in the space of probability measures.

*Proof.* We have

$$\begin{array}{rcl} \mathrm{P}_{t+s}(x, \mathrm{K}) \,=\, \mathbb{E}_x \mathbb{1}_\mathrm{K}(x_{t+s}) & = & \mathbb{E}_x \left[\mathbb{E}_x(\mathbb{1}_\mathrm{K}(x_{t+s}) \mid \mathcal{F}_t)\right] \\ & = & \mathbb{E}_x \mathrm{P}_s(x_t, \mathrm{K}) \,=\, \displaystyle\int_\mathrm{X} \mathrm{P}_t(x, \mathrm{d}y)\mathrm{P}_s(y, \mathrm{K}) \end{array}$$

where we used the fact that the law of $x_t$ under $\mathbb{P}_x$ coincides with $\mathrm{P}_t(x, \cdot)$. □

To each Markov process there correspond two families of linear operators acting in the spaces of bounded measurable functions $\mathrm{L}^\infty(\mathrm{X})$ and of probability measures $\mathcal{P}(\mathrm{X})$.

**Definition 42.** The **transition operator** $\mathrm{P}_t : \mathrm{L}^\infty(\mathrm{X}) \longrightarrow \mathrm{L}^\infty(\mathrm{X})$ of $(x_t)$ is defined for all $h \in \mathrm{L}^\infty(\mathrm{X}, \mathbf{R})$ by:

$$\begin{array}{rccc} \mathrm{P}_t h : & \mathrm{X} & \longrightarrow & \mathbf{R} \\ & x & \longmapsto & \displaystyle\int_{\mathrm{X}} h(y) \mathrm{P}_t(x, \mathrm{d}y). \end{array}$$

**Definition 43.** The **dual transition operator** $\mathrm{P}_t^* : \mathcal{P}(\mathrm{X}) \longrightarrow \mathcal{P}(\mathrm{X})$ of $(x_t)$ is defined for all $\mu \in \mathcal{P}(\mathrm{X})$ by:

$$\begin{array}{rccc} \mathrm{P}_t^* \mu : & \mathcal{B}_{\mathrm{X}} & \longrightarrow & \mathbf{R} \\ & \mathrm{K} & \longmapsto & \displaystyle\int_{\mathrm{X}} \mathrm{P}_t(x, \mathrm{K}) \mathrm{d}\mu(x). \end{array}$$

**Proposition 52** (Semigroup property)**.** *These two operators are semigroups.*

*Proof.* Use the Kolmogorov-Chapman relation. □

**Proposition 53** (Duality relation)**.** *They satisfy the following relation:*

$$\forall\, f \in \mathrm{L}^\infty(\mathrm{X}), \quad \forall\, \mu \in \mathcal{P}(\mathrm{X}) \ : \quad \langle \mathrm{P}_t f, \mu \rangle \,=\, \langle f, \mathrm{P}_t^* \mu \rangle \,.$$

*Proof.* Use the Fubini's theorem. □

**Proposition 54.** *The $\mathbb{P}_\lambda$-law of $x_t$ coincides with $\mathrm{P}_t^* \lambda$.*

**Corollary 55.**

$$\forall\, f \in \mathrm{L}^\infty(\mathrm{X}) \ : \quad \mathbb{E}_\lambda f(x_t) \,=\, \int_X \mathrm{P}_t f(x) \lambda(\mathrm{d}x).$$

**Definition 44.** A measure $\mu \in \mathcal{P}(X)$ is said to be **stationary** for $(x_t, \mathbb{P}_x)$ if:

$$\forall\, t \in \mathrm{T} \ : \quad \mathrm{P}_t^* \mu \,=\, \mu.$$

## B. **Noncommutative algebra results**

### B.1. **Exterior powers**

Let E be a real vector space of dimension $d$. For $1 \leqslant k \leqslant d$ we consider the $k$-fold exterior power of E, denoted $\wedge^k \mathrm{E}$, be the vector space of alternating $k$-linear forms on the dual space $\mathrm{E}^*$. We can identify this set with the set of formal expressions:

$$\sum_{i=1}^{m} c_i \left( u_1^{(i)} \wedge \ldots \wedge u_k^{(i)} \right),$$

with $m \in \mathbf{N}$, $c_i \in \mathbf{R}$ and $u_j^{(i)} \in \mathrm{E}$ if we do computations with the following conventions:

(1) $u_1 \wedge \ldots \wedge \left( u_j + u_j' \right) \wedge \ldots \wedge u_k \,=\, (u_1 \wedge \ldots \wedge u_j \wedge \ldots \wedge u_k) + \left( u_1 \wedge \ldots \wedge u_j' \wedge \ldots \wedge u_k \right)$,

(2) $u_1 \wedge \ldots \wedge c u_j \wedge \ldots \wedge u_k \,=\, c\,(u_1 \wedge \ldots \wedge u_j \wedge \ldots \wedge u_k)$,

(3) $\forall\, \sigma \in \mathfrak{S}_k \ : \quad u_{\sigma(1)} \wedge \ldots \wedge u_{\sigma(k)} \,=\, \mathrm{sign}(\pi)\, u_1 \wedge \ldots \wedge u_k$.

The elements in $\wedge^k E$ of the form $u_1 \wedge \ldots \wedge u_k$ are called **decomposable $k$ vectors**.
The following facts can be readily checked.

**Lemma 56** (Exterior powers of spaces and operators)**.** *Let $(e_1, \ldots, e_d)$ be a basis of* E.

(*i*) $(e_{i_1} \wedge \ldots \wedge e_{i_k})_{1 \leqslant i_1 < \ldots < i_k \leqslant d}$ *is a basis of* $\wedge^k \mathrm{E}$. *In particular,*

$$\dim(\wedge^k \mathrm{E}) \,=\, \binom{d}{k}.$$

(*ii*) $\forall\,(u_1,\ldots,u_k)\ :\quad u_1\wedge\ldots\wedge u_k\,=\,0\qquad\Longleftrightarrow\qquad u_1,\ldots,u_k$ *are linearly dependent.*

(*iii*) *Two sets* $(u_1,\ldots,u_k)$ *and* $(v_1,\ldots,v_k)$ *of* $k$ *linearly independent vectors in* $\mathrm{E}$ *have the same linear span if and only if there exists* $\lambda\in\mathbf{R}$ *such that:*

$$u_1\wedge\ldots\wedge u_k\,=\,\lambda(v_1\wedge\ldots\wedge v_k).$$

(*iv*) *If* $\mathrm{E}$ *has the scalar product* $\langle\cdot,\cdot\rangle$*, then the bilinear extension of*

$$\langle u_1\wedge\cdots\wedge u_k, v_1\wedge\cdots\wedge v_k\rangle\ :=\ \det(\langle u_i,v_j\rangle)_{k\times k},$$

*defines a scalar product in* $\wedge^k\mathrm{E}$*. In particular,*

$$\|u_1\wedge\cdots\wedge u_k\|\,=\,\sqrt{\det(\langle u_i,u_j\rangle)_{k\times k}}$$

*is the volume of the* $k$*-dimensional parallelepiped spanned by* $u_1,\ldots,u_k$*. Further, the generalized Hadamard's inequality holds: For* $1\leqslant p<k$*:*

$$\|u_1\wedge\cdots\wedge u_k\|\,\leqslant\,\|u_1\wedge\cdots\wedge u_p\|\,\|u_{p+1}\wedge\cdots\wedge u_k\|\,.$$

(*v*) *If* $A:\mathrm{E}\longrightarrow\mathrm{E}$ *is a linear operator, then:*

- *the linear extension of*

$$\wedge^k A(u_1\wedge\cdots\wedge u_k)\ :=\ Au_1\wedge\cdots\wedge Au_k$$

*defines a linear operator* $\wedge^k A$ *on* $\wedge^k\mathrm{E}$*, the* $k$*-fold exterior power of* $A$*, with*

$$\wedge^1 A\ =\ A,\qquad \wedge^d A\,=\,\det A\qquad\text{and}\qquad \wedge^k I\,=\,I.$$

- $\wedge^k A$ *has eigenvalues* $(\lambda_{i_1}\cdots\lambda_{i_k})_{1\leqslant i_1<\cdots<i_k\leqslant d}$*, where* $\lambda_1,\ldots,\lambda_d$ *are the eigenvalues of* $A$*,*
- *for all* $B$*:*

$$\wedge^k(AB)=(\wedge^k A)(\wedge^k B),\quad (\wedge^k A)^{-1}\,=\,\wedge^k A^{-1},\quad \wedge^k(cA)=c^k\wedge^k A,$$

- $\det(\wedge^k A)\,=\,(\det A)^{\binom{d-1}{k-1}}$,
- *If* $A$ *is an upper/lower triangular matrix in some basis then so is* $\wedge^k A$ *in the corresponding basis of (i).*

(*vi*) *Let* $E$ *have a scalar product and let* $\wedge^k E$ *be endowed with the scalar product introduced in part (iv). If* $U$ *is orthogonal, then* $\wedge^k U$ *is orthogonal. Further,*

$$\Big\langle\wedge^k A(u_1\wedge\ldots\wedge u_k), v_1\wedge\ldots\wedge v_k\Big\rangle\,=\,\det(\langle Au_i,v_j\rangle)_{k\times k},$$

*hence, with* $\big\|\wedge^k A\big\|$ *denoting the corresponding operator norm,*

$$\Big\|\wedge^k(AB)\Big\|\,\leqslant\,\Big\|\wedge^k A\Big\|\,\Big\|\wedge^k B\Big\|\qquad\text{and}\qquad(\wedge^k A)^*\,=\,\wedge^k A^*.$$

**Lemma 57** (Singular values of exterior power)**.** *Let* $A\in\mathcal{M}_d(\mathbf{R})$*, let* $A=VDU$ *be a singular value decomposition and let* $\sigma_1\geqslant\ldots\geqslant\sigma_d\geqslant 0$ *be the singular values of* $A$*. Then:*

(*i*) *a singular value decomposition of* $\wedge^k A$ *is:*

$$\wedge^k A\,=\,(\wedge^k V)(\wedge^k D)(\wedge^k U).$$

(*ii*) *for* $k \in [\![1, d]\!]$ :

- $\wedge^k D = \mathrm{Diag}(\sigma_{i_1} \cdots \sigma_{i_k},\ 1 \leqslant i_1 < \ldots < i_k \leqslant d)$,
- *the top singular value of* $\wedge^k A$ *is* $\sigma_1 \cdots \sigma_k$,
- *the smallest is* $\sigma_{d-k+1} \cdots \sigma_d$.

*Moreover, if* $A$ *is positive-definite (semi-definite) then so is* $\wedge^k A$.

(*iii*) *for all* $1 \leqslant k, m, k+m \leqslant d$ :

- $\left\|\wedge^k A\right\| = \sigma_1 \cdots \sigma_k \qquad \text{and} \qquad \left\|\wedge^d A\right\| = \sigma_1 \cdots \sigma_d = |\det A|$,
- $\left\|\wedge^{k+m} A\right\| \leqslant \left\|\wedge^k A\right\| \|\wedge^m A\|$,
- $\left\|\wedge^k A\right\| \leqslant \|A\|^k$,

*where* $\|\cdot\|$ *is the operator norm associated with the standard Euclidean norm in* $\mathbf{R}^d$.

B.2. **Proof of Furstenberg lemma**

- First point: assume that H is a compact subgroup of $\mathrm{SL}_d(\mathbf{R})$.

By compacity, H admits a right-invariant Haar probability measure $\gamma$, that is: $\gamma$ is a Borel probability measure on H such that

$$\forall\, A \in \mathrm{H}, \quad \forall\, K \in \mathcal{B}_{\mathrm{H}} \ : \quad \gamma(KA) = \gamma(K).$$

Hence, we define $\langle \cdot, \cdot \rangle'$ on $\mathbf{R}^d$ by:

$$\forall\, u, v \in \mathbf{R}^d \ : \quad \langle u, v \rangle' := \int_{\mathrm{H}} (Au, Av)\, \gamma(\mathrm{d}A),$$

where $(\cdot, \cdot)$ the standard inner product on $\mathbf{R}^d$. Using right-invariance of $\gamma$, we have that

$$\forall\, u, v \in \mathbf{R}^d, \quad \forall\, A \in \mathrm{H} \ : \quad \langle Au, Av \rangle' = \langle u, v \rangle'.$$

- Second point: assume that H is a non-compact.

Let us first state and prove the following useful lemma.

**Lemma 58.** *Let* $k \geqslant 1$ *and let* $(M_n)_n \in \mathrm{SL}_k(\mathbf{R})$ *such that*

$$\|M_n\| \ \xrightarrow[n+\infty]{} \ \infty.$$

*Then, on refining to a subsequence* $(M_{\varphi(n)})_n$, *there exist proper linear subspaces* $\mathrm{V}_1, \mathrm{V}_2 \subset \mathbf{R}^k$ *for which*

$$\forall\, v \notin \mathrm{V}_1 \ : \quad \mathrm{d}(M_{\varphi(n)} v, \mathrm{V}_2) \ \xrightarrow[n+\infty]{} \ 0.$$

*Proof.* Using the fact that $\det(M_n) = 1$ for all $n$, we can, without loss, pass to a subsequence with the property that for some fixed $1 \leqslant \ell < k$, we have

$$\frac{\sigma_\ell(M_n)}{\sigma_{\ell+1}(M_n)} \ \xrightarrow[n+\infty]{} \ +\infty. \tag{$*$}$$

Applying the Singular Value Decomposition to each $M_n$:

- let $\mathrm{V}_1^n$ be the unique $(k-\ell)$-dimensional subspace for which: $\|M_n\|_{\mathrm{V}_1^n} = \sigma_{\ell+1}(M_n)$,
- let $\mathrm{V}_2^n$ be the unique $\ell$-dimensional subspace for which: $\left\|M_n^{-1}\right\|_{\mathrm{V}_2^n} = (\sigma_\ell(M_n))^{-1}$.

Passing to a further subsequence, we can assume that the subspaces $\mathrm{V}_1^n, \mathrm{V}_2^n$ converge to subspaces $\mathrm{V}_1, \mathrm{V}_2$, respectively. The result now follows from $(*)$. □

Now, consider the set $\mathcal{G}$ of finite tuples of proper, nontrivial, distinct subspaces $(\mathrm{E}^1, \dots, \mathrm{E}^p)$ of $\mathbf{R}^d$ for which

$$\eta\left(\bigcup_{i=1}^{p} \mathrm{E}^i\right) \,=\, 1.$$

Applying the previous lemma to a sequence $(M_n) \in \mathrm{H}^{\mathbf{N}}$ with $\|M_n\| \longrightarrow \infty$, note that the pair $(\mathrm{V}_1, \mathrm{V}_2)$ so-obtained is such a tuple. If $(\mathrm{E}_1, \dots, \mathrm{E}_p)$, $(\mathrm{F}_1, \dots, \mathrm{F}_q)$ are two such tuples, let us write:

$$(\mathrm{E}^i) \,\leqslant\, (\mathrm{F}^i) \qquad \Longleftrightarrow \qquad \bigcup_i \mathrm{E}_i \,\subset\, \bigcup_i \mathrm{F}_i.$$

Note that $\leqslant$ is a partial order on $\mathcal{G}$. We say that two tuples $(\mathrm{E}_1, \dots, \mathrm{E}_p)$, $(\mathrm{F}_1, \dots, \mathrm{F}_q)$ in $\mathcal{G}$ are equivalent up to relabeling if:

$$p \,=\, q \qquad \text{and} \qquad \exists\, \pi \in \mathfrak{S}_p \,:\quad \forall\, j \in [\![1, p]\!] \,:\quad \mathrm{F}^j \,=\, \mathrm{E}^{\pi(j)}.$$

Let us state and prove a series of Lemmas which the second point.

**Lemma 59.** *There is a unique tuple $(\mathrm{E}_1, \dots, \mathrm{E}_p)$ (up to relabeling) of distinct, proper and non-trivial linear subspaces of $\mathbf{R}^d$ minimal with respect to the partial order $\leqslant$ on $\mathcal{G}$, and such that:*

$$\forall\, A \in \mathrm{H}_\eta, \quad \exists\, \pi \,=\, \pi_A \in \mathfrak{S}_p, \quad \forall\, i \in [\![1, p]\!] \,:\quad A\mathrm{E}^i \,=\, \mathrm{E}^{\pi(i)}.$$

*Proof.* Straightforward. □

The minimal tuple $(E^i)$ therefore satisfies conditions $(i)$-$(ii)$ in the Lemma 41. Item $(iii)$ is verified below.

**Lemma 60.** *For each $i \in [\![1, p]\!]$, there is an inner product $\langle \cdot, \cdot \rangle^i$ on $\mathrm{E}^i$ such that:*

$$\forall\, A \in \mathrm{H} \,:\quad A : (\mathrm{E}^i, \langle \cdot, \cdot \rangle^i) \,\longrightarrow\, (\mathrm{E}^{\pi(i)}, \langle \cdot, \cdot \rangle^{\pi(i)}) \text{ is a conformal mapping,}$$

*where $\pi \,=\, \pi_A \in \mathfrak{S}_p$.*

*Proof.* We denote $\widetilde{\mathrm{H}} \subset \mathrm{H}$ a closed, normal subgroup of finite index

$$\widetilde{\mathrm{H}} \,=\, \widetilde{\mathrm{H}}_\eta \,=\, \left\{A \in \mathrm{H}_\eta \,:\, A\mathrm{E}^i \,=\, \mathrm{E}^i\right\}.$$

The quotient group $\mathrm{H}/\widetilde{\mathrm{H}}$ is naturally isomorphic to a subgroup of $\mathfrak{S}_p$.

Let us assume for the moment that $\mathfrak{S}_p$ acts transitively on $[\![1, p]\!]$. Fix an arbitrary $i \in [\![1, p]\!]$ and form

$$\check{\mathrm{H}}^{(i)} \,=\, \left\{\det\left(A_{|\mathrm{E}^i}\right)^{-\frac{1}{\dim(\mathrm{E}^i)}} A_{|\mathrm{E}^i},\; A \in \widetilde{\mathrm{H}}\right\}.$$

Note that linear operators in $\check{\mathrm{H}}^{(i)}$ preserve the measure $\eta_{|\mathrm{E}^i}$. Since any $A \in \widetilde{\mathrm{H}}$ maps $\mathrm{E}^i$ into itself, we can think of $\check{\mathrm{H}}^{(i)}$ as a subgroup of $\mathrm{SL}_{\dim \mathrm{E}^i}(\mathbf{R})$ on identifying $\mathrm{E}^i$ with $\mathbf{R}^{\dim \mathrm{E}^i}$.

1. We have that $\check{\mathrm{H}}^{(i)}$ is compact. Indeed, if it is not compact, then by Lemma 58 there are proper linear subspaces $\mathrm{W}^1, \mathrm{W}^2 \subset \mathrm{E}^i$ for which $\eta\left(\mathrm{W}^1 \cup \mathrm{W}^2\right) \,=\, \eta(\mathrm{E}^i)$. This contradicts minimality of $(\mathrm{E}_1, \dots, \mathrm{E}_p)$ as in item $(i)$.

2. It now follows from $(a)$ that there exists an inner product $\langle \cdot, \cdot \rangle^i$ on $\mathrm{E}^i$ with respect to which $\check{\mathrm{H}}^{(i)}$ acts isometrically. Equivalently, linear operators of the form $A_{|\mathrm{E}^i}, A \in \widetilde{\mathrm{H}}$ act conformally with respect to $\langle \cdot, \cdot \rangle^i$.

3. Since $\mathfrak{S}_p = \mathrm{H}/\widetilde{\mathrm{H}}$ acts transitively on $[\![1,p]\!]$:

$$\forall\, j \in [\![1,p]\!],\ j \neq i, \quad \exists\, M \in \mathrm{H} \ : \quad M\mathrm{E}^i = \mathrm{E}^j$$

and we define $\langle \cdot, \cdot \rangle^j$ as:

$$\forall\, v, w \in \mathrm{E}^j \ : \quad \langle v, w \rangle^j = \left\langle M^{-1}v, M^{-1}w \right\rangle^i .$$

4. This definition is independent of $M$ : if $N\mathrm{E}^i = \mathrm{E}^j$ for some other $N \in \mathrm{H}$, then

$$\forall\, v, w \in \mathrm{E}^j \ : \quad \left\langle N^{-1}v, N^{-1}w \right\rangle^i = \left\langle M^{-1}v, M^{-1}w \right\rangle^i = \langle v, w \rangle^j .$$

By a similar computation, one checks that if $A \in \mathrm{H}$ maps $A\mathrm{E}^i = \mathrm{E}^j$, then $A$ is conformal with respect to the inner products $\langle \cdot, \cdot \rangle^i, \langle \cdot, \cdot \rangle^j$, respectively.

This completes the proof when $\mathfrak{S} \cong \mathrm{H}/\widetilde{\mathrm{H}}$ acts transitively on $[\![1,p]\!]$.

Let us now address the situation when $\mathfrak{S}_p$ does not act transitively on $[\![1,p]\!]$.

1. By a standard argument there is a unique partition $(\mathcal{P}_l)_{1 \leqslant l \leqslant k}$ of $[\![1,p]\!]$, such that:

$$\mathfrak{S}_p \mathcal{P}_l = \mathcal{P}_l, \qquad \text{and} \qquad \mathfrak{S}_p \text{ acts transitively on } \mathcal{P}_l.$$

2. For each $\mathcal{P}_l$, repeat the construction of $\langle \cdot, \cdot \rangle^i$ for some fixed arbitrary $i \in \mathcal{P}_l$, and then define $\langle \cdot, \cdot \rangle^j, j \in \mathcal{P}_l, j \neq i$ as in previously for some arbitrary $M \in H$ sending $M\mathrm{E}^i = \mathrm{E}^j$ (such an $M$ exists since $\mathfrak{S}$ acts transitively on $\mathcal{P}_l$ by construction).

3. The result now follows from the previous arguments, since for all $A \in H$, we can have $A\mathrm{E}^i = \mathrm{E}^j$ only if $i, j$ belong to the same $\mathcal{P}_l$ for some $1 \leqslant l \leqslant k$.

□

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